Growth with Endogenous Industry Dynamics

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Abstract

This paper develops a Schumpeterian growth model in which endogenous industry lifecycles shape aggregate growth. Potential entrants choose whether to join an existing industry or create a new one. Industries begin as monopolies, attract followers into a patent race, and gradually mature as innovation slows. Individual industries follow non-stationary paths, yet a stationary cross-section sustains steady aggregate growth as new industries continually replace maturing ones. This cross-section delivers an explicit decomposition of growth into variety creation, frontier innovation, and catch-up innovation. Because entrants cannot fully appropriate the knowledge a new industry adds, the decentralized economy creates too few new industries relative to the constrained optimum. Calibrated to U.S. data, the model shows that the effectiveness of innovation policy depends on how the cross-section of industries responds to this entry margin.

Keywords: Schumpeterian growth; Endogenous industry dynamics; Industry lifecycle; Variety creation; Innovation policy

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1 Introduction

Aggregate productivity growth has remained relatively stable over long horizons, even though the sectors driving it rotate over time (Ferguson and Wascher, 2004; Sarte and Taylor, 2025). This rotation reflects life-cycle patterns at the industry level, in which young industries exhibit high entry and rapid innovation while mature industries see declining entry and more incremental innovation (Gort and Klepper, 1982; Abernathy and Utterback, 1978; Klepper, 1996). As industries mature and their innovation slows, sustained aggregate growth requires a continuing flow of new, high-growth industries. Firms drive this process through entry and innovation decisions that vary with an industry's stage of development and in turn shape industry dynamics. The interaction between firm decisions and industry life-cycles is therefore central both to explaining how aggregate growth stays steady despite the rise and decline of individual industries and to evaluating innovation policy.

In this paper, I develop and quantify a step-by-step Schumpeterian growth model in which the set of active industries is endogenous. The key new margin is that potential entrants choose whether to join an existing industry or pay a sunk cost to create a new one. Industries are created by entrants, evolve as followers join and competition intensifies, and eventually exit due to obsolescence. Within each industry, the intensity of competition is itself endogenous, determined by the number of rivals and the size of technology gaps. This intensity shapes both incumbent innovation incentives and the attractiveness of the industry to new entrants. The entry decision governs the rate at which new industries appear, the cross-sectional distribution of industries over life-cycle states, and the long-run growth rate.

The paper makes three contributions. First, it endogenizes industry life-cycles in a Schumpeterian growth framework and shows how they shape aggregate growth. The entry mechanism generates the transition from new monopoly industries to mature oligopolies as an equilibrium outcome, and although individual industries are non-stationary, a stationary cross-sectional distribution over life-cycle states sustains steady growth and yields an explicit decomposition of it across the life-cycle. Second, it characterizes the constrained inefficiency of decentralized industry creation. Because entrants cannot fully appropriate the knowledge a new industry adds, the economy undertakes too little variety creation relative to the constrained optimum. Third, the model introduces a policy margin absent from fixed-industry frameworks, the allocation of entrepreneurial entry between incumbent and new industries. Because this margin is undervalued in equilibrium, equal-cost experiments show that subsidies to variety creation can raise growth and welfare by more than subsidies to incumbent R&D.

New industries begin as small monopolies with a one-step lead over a competitive fringe.

As they prove profitable, they attract followers, transition into oligopoly, and experience high innovation rates as leaders and followers compete in a patent race. Over time, leaders pull ahead, followers catch up less frequently, and the incentives for further entry and follower innovation decline. Mature industries become larger but less dynamic, and joining them becomes less attractive for new entrepreneurs. At that point, potential entrants are more likely to bear the cost of creating new industries instead.

These non-stationary paths combine into a stationary cross-section, and the aggregate growth rate decomposes into three components. Variety creation comes from the birth of new industries, frontier innovation from leaders pushing ahead within monopolies, and catch-up innovation from followers closing gaps in oligopolies, each weighted by the stationary share of industries in the corresponding state. Because these weights are the life-cycle shares, the cross-section maps directly to the growth rate, which clarifies how stable long-run growth can coexist with front-loaded innovation and declining business dynamism at the industry level.

The same entry decision that drives these dynamics also creates a wedge between private and social incentives to create new industries. A new industry starts one step above the public knowledge level, so its creation expands the economy's knowledge stock in a way that entrants cannot fully appropriate. Creative destruction amplifies this wedge, because the arrival of future industries makes private firms discount profits at a rate above the social rate. The resource cost of market creation partly offsets it, since the planner values that cost at a shadow price that reflects the aggregate resource constraint. A fourth force operates through general equilibrium, as a change in the entry margin shifts the whole distribution of industries and feeds back into firm values and innovation incentives. In the calibrated economy, the knowledge and creative-destruction forces dominate the resource cost, and this general-equilibrium adjustment does not reverse the ranking, so the decentralized economy creates fewer new industries than the constrained planner would choose. The effectiveness of innovation subsidies therefore depends on how the cross-section responds to the entry margin.

I discipline the magnitudes by calibrating the model to U.S. data from 2000–2019. In the calibrated economy, too many entrants join existing oligopolies to capture appropriable business-stealing rents, and too few bear the fixed costs of opening new markets. Comparative statics reveal a U-shaped relationship between startup entry and growth, reflecting the tension between business-stealing and variety-creation effects. The relationship between competition and growth is inverted-U-shaped, as tougher within-industry competition first spurs incumbent innovation and redirects entrants toward new industries and later erodes the rents that sustain growth. In equal-cost policy experiments, a subsidy to variety creation produces larger gains in growth and welfare than a subsidy to incumbent R&D, even though both work mainly through incumbent innovation. By drawing entrants toward new industries, the variety-creation subsidy relieves entry pressure on mature incumbents and raises their innovation incentives, rather than sharply increasing the rate of variety creation.

Related Literature This paper connects four literatures that have developed largely in parallel. The first studies within-industry Schumpeterian competition and quality-ladder innovation. The second studies growth through product creation and technological turnover. The third documents life-cycle regularities at the industry and product level. The fourth studies innovation policy in endogenous-growth and industry-dynamics environments. My contribution is to bring these strands together by endogenizing industry life-cycles in a Schumpeterian growth framework, so that the life-cycle structure shapes aggregate growth, leaves decentralized industry creation inefficient, and opens a policy margin that fixed-industry frameworks cannot produce.

A large literature studies within-industry innovation races and market structure in Schumpeterian growth models. Aghion et al. (2001) introduce the step-by-step framework in which firms' innovation incentives depend on their position relative to rivals. Akcigit and Ates (2021, 2023) extend this framework to match ten facts on declining U.S. business dynamism and identify declining knowledge diffusion as the dominant driver. Liu, Mian and Sufi (2022) use a similar structure to show that low interest rates raise market power and reduce productivity growth. Olmstead-Rumsey (2022) studies how declining laggard innovativeness drives rising concentration and the productivity slowdown. Cavenaile, Celik and Tian (2025) develop an oligopolistic Schumpeterian model with an endogenously determined number of large firms within each industry and strategic innovation choices, with implications for markups and business dynamism. These models generate rich within-industry dynamics, including escape-competition effects, endogenous technology gaps, and leadership turnover. However, their key entry margins operate within existing markets rather than between incumbent industries and entirely new industries. As a result, the rate at which the economy creates new markets is not an equilibrium object in their policy analysis.

A parallel literature studies growth through product creation and technological turnover. Romer (1990) introduced the expanding-variety growth model. Akcigit and Kerr (2018) distinguish internal innovation that improves existing products from external innovation that creates new product lines and find that entrants and small firms contribute disproportionately to high-impact external innovations. Garcia-Macia, Hsieh and Klenow (2019) decompose U.S. productivity growth and show that incumbent own-product improvements account for the largest share, a decomposition I return to in the policy analysis. In a related endogenous-growth model of technological turnover, Aghion et al. (2026) have entrants introduce new technologies and incumbents develop them incrementally, sustaining long-run growth through periodic replacement. Ribeiro (2026) studies growth with continually emerging technology vintages and a stationary distribution of research effort across technologies of different ages. But his central margin is the allocation of R&D across new and old technologies, whereas this paper focuses on the allocation of entrepreneurial entry across incumbent and new industries. Most existing models of product creation and technological turnover do not jointly model the gradual transition from monopoly to competitive oligopoly, a front-loaded innovation pattern that declines over a life-cycle, and a stationary cross-sectional distribution over industry life-cycle states that maps directly into aggregate growth. This paper brings these elements together.

A third literature documents recurring life-cycle regularities in industries and product markets. Gort and Klepper (1982) and Agarwal (1998) document high entry early in the life-cycle and later consolidation, while Klepper (1996) emphasizes declining product innovation and stabilizing market shares as industries mature. Abernathy and Utterback (1978) show that product innovation is front-loaded and gradually gives way to process improvements. Argente, Lee and Moreira (2024) document that individual products peak quickly and then decline as newer varieties capture market share. Argente et al. (2025*b*) combine patent and product data and document patterns consistent with strategic patenting by market leaders, which may impede rival product introduction. Jovanovic and Wang (2025) develop a model of industry evolution in which diffusion shapes innovation incentives and industry growth over the life-cycle, and calibrate it to 18 U.S. industries, including automobiles and personal computers. My model generates these industry-level regularities endogenously from the interaction of step-by-step innovation and entry decisions.

A fourth literature studies innovation policy in endogenous-growth and industry-dynamics environments. Akcigit, Hanley and Stantcheva (2022) derive optimal nonlinear R&D subsidies with heterogeneous firms but do not distinguish the variety-creation margin from incumbent R&D. Atkeson and Burstein (2019) evaluate uniform innovation subsidies and find modest scope for raising productivity. These papers do not study how subsidy design interacts with the composition of young and mature industries through endogenous industry creation. Cavenaile et al. (2024) are especially relevant because they study the policy implications of industry life-cycles in general equilibrium with endogenous innovation and oligopolistic competition following technological breakthroughs. My paper differs by embedding a step-by-step Schumpeterian patent-race structure in a model with an explicit entry margin between joining incumbent industries and founding new ones. The stationary distribution of industries across life-cycle states and the effects of different subsidy instruments are therefore jointly determined in equilibrium.

The rest of the paper is organized as follows. Section 2 reviews stylized facts on industry life-cycles and sectoral contributions to aggregate growth. Section 3 lays out the model. Section 4 characterizes the stationary Markov-perfect equilibrium. Section 5 derives the main theoretical results, a decomposition of aggregate growth, a constrained efficiency analysis, and a characterization of industry life-cycle dynamics. Section 6 presents the quantitative analysis and policy implications. Section 7 concludes.

2 Motivating Facts

This section reviews four stylized facts about industry dynamics and aggregate productivity growth. Each fact highlights a dimension of the tension between non-stationary industry evolution and stable aggregate growth, and each motivates a specific feature of the model developed in Section 3.

Stylized Fact 1: Aggregate productivity growth is stable, but the industries driving it rotate over time.

Over long horizons, aggregate productivity growth in advanced economies has been relatively stable. The industrial sources of that growth have not been stable. In the United States, the leading contributors have shifted repeatedly, from durable goods manufacturing to IT-producing industries and then toward IT-using services (Oliner, Sichel and Stiroh, 2007; Jorgenson, Ho and Stiroh, 2008). Sectoral reallocation continues in more recent data (Sarte and Taylor, 2025; Hobijn et al., 2025). These episodes are part of a broader pattern of recurring productivity surges associated with different technological and organizational forces (Ferguson and Wascher, 2004). If individual industries slow down as they mature, steady aggregate growth requires that new high-growth industries continually replace them. This replacement process motivates the central margin of the model, in which entrepreneurs endogenously create new industries. Individual industries then follow non-stationary life-cycle paths, but the cross-sectional distribution over life-cycle states remains stationary and delivers steady aggregate growth.

Stylized Fact 2: Industry-level output rises and prices fall.

At the industry level, Agarwal (1998), building on the diffusion patterns documented by Gort and Klepper (1982), shows that industry evolution is typically associated with rising quantities and falling prices across a wide range of markets. In the model, surviving industries grow in quantity and fall in relative price as entry adds firms and innovation raises frontier technology.

Stylized Fact 3: Entry rates decline as industries mature.

Gort and Klepper (1982) and Agarwal (1998) document that, across many product markets, the number of active firms rises strongly in early stages and then levels off or declines. Using a larger set of product markets with explicit entry and exit data, Agarwal and Gort (1996) show that industry entry rates peak in early growth stages and fall over time. Several mechanisms may amplify this decline. Argente et al. (2025*a*) highlight costly demand-side frictions in firm growth, while Argente et al. (2025*b*) document patterns consistent with strategic patenting by market leaders, suggesting an additional barrier to entry in mature markets. In the model, expected profits for new followers fall as an industry matures, so entry into a given industry declines endogenously. As incumbent industries become less attractive, potential entrants face stronger incentives to bear the sunk cost of founding a new market.[1]

Stylized Fact 4: Innovation is front-loaded in the industry life-cycle.

Abernathy and Utterback (1978) argue that new industries begin in a "fluid" phase with frequent major product innovations, after which a dominant design emerges and innovation shifts toward incremental process improvements. Klepper (1996) and Agarwal (1998) document similar trajectories, with patenting activity rising in the early years of a product market and declining in later stages. Jovanovic and Wang (2025) formalize these observations in a model of industry evolution calibrated to U.S. industries, including automobiles and personal computers. Front-loaded innovation ties aggregate growth to a continuous supply of young, high-innovation industries. If industry creation slows, the cross-section of industries ages and aggregate innovation declines even if within-industry behavior is unchanged. In the model, innovation rates decline endogenously over the industry life-cycle.

These four stylized facts point to a growth process in which aggregate regularities emerge from the interaction of many industries at different life-cycle stages, rather than from a representative sector. They also raise questions that the evidence alone cannot answer. Given that each industry eventually slows down, how does the continuous creation of new industries sustain aggregate growth, and does the decentralized economy get the rate of industry creation right? The model developed in Section 3 provides a framework for answering these questions.

[1] To keep the analysis tractable, I do not model the late-stage decline in the number of firms. A standard extension with fixed operating costs could generate such a shakeout without changing the main mechanisms.

3 Model

This section presents a continuous-time Schumpeterian growth model in which the set of active industries is endogenous. The model builds on step-by-step innovation frameworks (Akcigit and Ates, 2021, 2023; Liu, Mian and Sufi, 2022; Olmstead-Rumsey, 2022; Cavenaile, Celik and Tian, 2025) but departs from them in a key respect. Rather than treating the set of product lines as fixed, it allows industries to be created by entrepreneurs and to exit over time, generating *endogenous industry life-cycles.* Industries are born when entrepreneurs open new product markets, evolve through phases of monopoly and oligopolistic competition as followers enter and innovate, and eventually become obsolete. The central strategic tension arises from the entry decision that *potential entrants* face. They can enter an existing industry to compete for market share, or they can create an entirely new industry and obtain a temporary monopoly position. This choice between intensifying competition within existing markets and expanding the set of markets governs both aggregate growth and the structural evolution of the economy. The model reproduces the four stylized facts of Section 2 as equilibrium outcomes.

I first describe households and the final-good technology, then the structure of industries, and finally the innovation, entry, and exit processes. Where an assumption plays a specific role in supporting the stationary equilibrium or the growth decomposition derived in later sections, I note that role when the assumption is introduced.

3.1 Household

Time is continuous and indexed by $t \in [0, \infty)$. The economy is populated by a representative household that consumes, saves, and supplies labor. Production takes place in a continuum of industries indexed by $i \in [0, N_t]$, where the mass N_t changes over time as new industries are created and existing ones exit. The household has logarithmic preferences over a final consumption good and maximizes lifetime utility

$$U_t \;=\; \int_t^{\infty} e^{-\rho(s-t)} \log C_s \, ds, \tag{1}$$

where $\rho > 0$ is the rate of time preference and C_t denotes consumption at time t. Log preferences imply a closed-form welfare expression used in Section 5. The household owns aggregate wealth A_t, equal in equilibrium to the total market value of incumbent firms, and supplies one unit of labor inelastically. Its budget constraint is

$$C_t + \dot{A}_t \;=\; w_t + r_t A_t, \tag{2}$$

where w_t is the wage and r_t is the interest rate.

The final consumption good, which serves as the numeraire, is produced with a Cobb-Douglas aggregator over industry outputs:

$$Y_t \;=\; N_t \exp\left(\frac{1}{N_t}\int_0^{N_t} \ln Y_{it}\, di\right), \tag{3}$$

where N_t is the mass of industries at time t and Y_{it} is the output of industry i. The aggregator implies that each industry receives the same revenue flow $\bar{Y}_t \equiv Y_t/N_t$. Firm profits can therefore be written as functions only of within-industry state variables. This allows the aggregate economy to be summarized by a tractable cross-sectional distribution over industry states, characterized in Section 4.

The exponential term is a geometric average of industry outputs and therefore measures output per industry. The leading N_t converts this average into an aggregate. This normalization implies that a larger mass of industries does not mechanically raise aggregate output when aggregate inputs and the cross-sectional distribution of industry technologies are held fixed. The aggregator therefore creates no mechanical love-of-variety effect from a larger mass of industries. New industries still reallocate revenue away from incumbents through $\bar{Y}_t = Y_t/N_t$, but they contribute to aggregate growth only through the technologies they embody.

3.2 Industries and Production

Unless noted otherwise, I omit time subscripts for brevity. Each industry i contains a finite number of firms, collected in a set $\mathcal{F}_i$, that differ in their technology levels and therefore in their productivity and equilibrium markups. If an industry has only one firm, it is a *monopoly*. If there are multiple firms, the industry is an *oligopoly* with one *leader* (the firm with the highest technology level) and a set of *followers* that share a common (lower) technology level. The assumption that all followers share a common technology level keeps the within-industry state to two dimensions, the technology gap m and the follower count n. It also ensures that catch-up innovation maps directly into the evolution of the lowest technology tier used in the growth decomposition of Section 5. Firms produce differentiated varieties within the industry and compete in prices.

Within-industry demand and production. Industry i's output is a CES aggregate of the individual varieties produced by firms $j \in \mathcal{F}_i$:

$$Y_i \;=\; \left(\sum_{j\in\mathcal{F}_i} y_{ji}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1,$$

where y_{ji} is the quantity of a firm j's variety and σ is the elasticity of substitution across varieties within an industry. Industry demand for Y_i comes from the final-good sector.

Each firm j combines labor with its technology to produce:

$$y_{ji} \;=\; q_{ji}\, l_{ji}, \qquad q_{ji} \;=\; \lambda^{s_{ji}},$$

where l_{ji} is labor input, $s_{ji} \in \mathbb{R}_+$ is the technology level, and $\lambda > 1$ is the step size of the quality ladder. A one-step increase in s_{ji} raises firm j's productivity by a factor λ. Throughout the paper, I use *technology gap* to refer to the difference in technology levels, measured in steps on the quality ladder, between a firm and its closest competitors. Firms take wages and the demand system as given and set prices to maximize profits. For a given industry state (number of firms and their technology levels), price competition uniquely determines equilibrium markups and profits.

Competitive fringe in new industries. The economy features a public knowledge stock, denoted $s_t^K \in \mathbb{R}_+$, defined as the average across industries of the lowest technology level among active producers. This object measures the widely available knowledge that new entrants can draw on when designing a new product. When a new industry is created at time t, it starts with a monopolist whose technology is one step ahead of the public knowledge, $s_{ji} = s_t^K + 1$, and with a competitive fringe of firms whose technology equals s_t^K, the level of the knowledge stock on the creation date. Fringe firms do not innovate, so their technology remains at this level as s^K continues to rise. Each fringe firm can produce a perfect substitute for the monopolist's variety. In equilibrium, the monopolist sets a limit price equal to the fringe's unit cost, so fringe firms do not produce. The fringe therefore does not enter the computation of s^K. The fringe serves two roles. Economically, it captures the early exploratory phase of a new product line, in which many unsuccessful imitators can enter quickly using widely available technologies. Technically, the fringe ensures the monopolist faces a finite price elasticity and therefore a well-defined markup, as formalized below.

3.3 Innovation, Industry Creation, and Destruction

Firms invest in R&D to improve their technology. Innovation is uncertain and arrives at Poisson rates that depend on firms' R&D choices. At the same time, new potential entrants arrive and decide whether to join existing industries or create new ones. Industries exit exogenously. This subsection describes these processes and the implied channels of creative destruction and knowledge diffusion.

Innovation and R&D costs. Consider a firm that chooses an innovation intensity $x \geq 0$. To achieve this rate, the firm must incur a flow R&D cost $R(x)\bar{Y}$ in units of the final good, where

$$R(x) \;=\; \frac{\alpha}{\psi}x^{\psi}, \qquad \alpha > 0,\ \psi > 1. \tag{4}$$

The parameter ψ governs the curvature of the R&D technology. The scale parameter α controls the overall costliness of innovation. Denominating the cost in units of $\bar{Y}$ scales R&D spending with the size of the industry's market and allows innovation incentives to be stationary after firm values are normalized by industry revenue.

Conditional on choosing x, innovations arrive according to a Poisson process with intensity x. Each successful innovation raises the firm's technology index s_{ji} by one step, $s_{ji} \;\to\; s_{ji}+1$, and thus, $q_{ji} \;\to\; \lambda q_{ji}$. For followers in an oligopoly, I assume that knowledge is shared instantaneously among all followers. When any follower innovates, the technology level of all followers in that industry increases by one step. Thus, from the perspective of an individual follower, the effective innovation intensity includes both its own effort and the efforts of its peers. This sharing sustains the common follower technology level assumed in Section 3.2. It captures the idea that incremental improvements are easily codified and diffuse among firms using similar, relatively mature technologies.[2]

Potential entrants and industry creation. In each existing industry, a potential entrant arrives at an exogenous Poisson rate $z > 0$. This arrival rate governs the flow of entry opportunities, while the equilibrium allocation of those opportunities between incumbent entry and new-industry creation is endogenous. Upon arrival, the potential entrant chooses between:

[2] I abstract from exogenous catch-up shocks and from sudden leapfrogging within incumbent industries. In many related step-by-step frameworks, such assumptions are used to keep technology gaps from becoming too persistent or to generate sufficient turnover. In this framework, they are not needed. Catch-up innovation is endogenous and gradual, arising from followers' R&D choices, while large gaps and crowded follower tiers endogenously depress follower values, innovation, and entry into mature industries. Stationarity is instead sustained by the continual creation of new industries, which replenishes the cross-section with young, high-growth industries and prevents the economy from being dominated by mature, low-innovation ones.

1. *Joining the industry where it arrived*: the entrant adopts the lowest technology level among active producers in that industry (the followers' level in an oligopoly, or the monopolist's level in a monopoly), becoming either a new follower or a neck-and-neck competitor. This reflects the idea that older technologies are easier to access and that entry into established markets typically occurs at the lower end of the quality spectrum.

2. *Creating a new industry*: the entrant starts a fresh industry as a monopolist at technology level $s^K + 1$, one step ahead of the current public knowledge stock s^K. Creating a new industry requires paying a sunk cost $\kappa\bar{Y}$ in units of the final good. The cost κ is an i.i.d. draw from a distribution G with support $[\underline{\kappa}, \overline{\kappa}]$, $\underline{\kappa} \geq 0$, which the entrant observes upon arrival. A lower draw κ represents a more promising or more easily implementable idea for a new product line.

After observing the state of the industry where it arrived and its cost draw κ, the potential entrant compares the value of joining that industry with the value of paying $\kappa\bar{Y}$ to found a new industry that starts with a temporary monopoly position. This entry decision jointly determines the number of firms in each industry and the rate at which new industries appear. It therefore governs the balance between intensifying competition within existing markets and expanding the set of markets. Because a new industry is born at $s^K + 1$, the public knowledge stock and the rate of industry creation are linked. As industries innovate and followers catch up, s^K gradually increases, and each successive new industry starts from a higher technological base. This link between industry creation and the knowledge stock is also the source of the knowledge externality analyzed in Section 5. Private entrants cannot appropriate the full productivity gain that a new industry confers on the economy.

Industry exit. Each active industry exits the economy at an exogenous Poisson rate $\delta > 0$. When an industry exits, all firms in that industry disappear and their products cease to be available in the final-good aggregator. Industry exit captures obsolescence of product lines due to shifts in tastes, technologies, or regulations that are not explicitly modeled. Together with the continual creation of new industries, exogenous exit prevents the cross-section from drifting toward arbitrarily old, low-innovation industries and helps sustain the stationary distribution characterized in Section 4.

Before turning to the dynamic equilibrium, it is useful to note that simple parameter restrictions nest two familiar benchmark environments. Setting $z = 0$ and $\delta = 0$ shuts down industry creation and destruction. The mass of industries is then fixed and the model reduces to a step-by-step quality-ladder framework in which growth is driven solely by within-industry innovation. If I instead set $\alpha \to \infty$ so that incumbent firms do not invest in R&D,

while keeping $z > 0$, long-run growth is driven solely by the creation of new industries, each born one step above the rising knowledge stock, as in expanding-variety models. The baseline environment studied below maintains both $\alpha < \infty$ and $z > 0$, so that aggregate growth reflects a combination of quality improvements within industries and the creation of new industries over time.

4 Equilibrium

This section characterizes a stationary and symmetric Markov-perfect equilibrium. I first establish the industry states and profit structure implied by the model primitives. I then describe the value functions and innovation decisions of monopolists, leaders, followers, and potential entrants. Finally, I define the equilibrium, express the rate of variety creation in terms of the stationary distribution of industries over life-cycle states, and derive the technology dynamics that enter the growth decomposition in Section 5.

4.1 Industry States and Profit Structure

Equilibrium prices, quantities, and profits within an industry are functions of a small set of state variables. To summarize the state of an industry, I adopt the following notation. A superscript M refers to monopolies or monopolist-specific variables. $\hat{s}$ denotes the technology gap between a monopolist and the fringe. For example, $V_{\hat{s}}^M$ is the value of a monopolist with gap $\hat{s}$ over the fringe. A subscript (m, n) refers to an oligopoly in which the leader's technology gap over the followers is m and there are n followers. I call such an industry an (m, n)-industry. For example, Π_{mn} denotes the leader's profit in an (m, n)-industry. A subscript $-mn$ refers to a representative follower in an (m, n)-industry.

Given the Cobb-Douglas aggregation in the final-good sector, each active industry earns the same revenue flow $\bar{Y} \equiv \frac{Y}{N}$ in equilibrium, and I use $\bar{Y}$ to scale firm-level variables. The equilibrium firm profits are proportional to industry revenue and expressed as $\Pi = \pi\bar{Y}$, where $\pi \in (0, 1)$ is a normalized profit. The industry price is defined as $P_i \equiv \frac{\sum_{j\in\mathcal{F}_i} p_{ji}y_{ji}}{Y_i}$, where p_{ji} is the price charged by firm $j \in \mathcal{F}_i$. Profits, output, and prices satisfy the following properties in equilibrium.

Lemma 4.1 (Properties of firm profits and industry output and price)**.** *Consider a monopoly industry born at time τ and currently with technology gap $\hat{s}$ over the fringe, and an oligopoly with gap m, n followers, and follower technology level s_f.*

1. *The monopolist's normalized profit is $\pi_{\hat{s}}^M = 1 - \lambda^{-\hat{s}}$. Hence $\pi_{\hat{s}}^M \nearrow 1$, and the one-step profit gain $\pi_{\hat{s}+1}^M - \pi_{\hat{s}}^M$ decreases to zero as $\hat{s} \to \infty$.*

2. *The oligopoly leader and follower's normalized profits, π_{mn} and π_{-mn}, depend only on m and n. As $m \to \infty$, $\pi_{mn} \nearrow 1$ and $\pi_{-mn} \searrow 0$; as $n \to \infty$, $\pi_{mn} \searrow 0$ and $\pi_{-mn} \searrow 0$.*

3. *The wage-adjusted industry price P_i/w depends only on s^K_τ in the monopoly case and (m, n, s_f) in the oligopoly case. $P_i/w \searrow 0$ as $s^K_\tau \to \infty$ for a monopoly, and as $m \to \infty$, $n \to \infty$, or $s_f \to \infty$ for an oligopoly. Industry output satisfies $Y_i = \bar{Y}/P_i$ and, for given $\bar{Y}/w$, $Y_i \nearrow \infty$ in each of these limits.*

Lemma 4.1 establishes the profit and price properties that shape innovation incentives over the industry life cycle. A monopolist gains from widening its technology gap over the competitive fringe, since $\pi^M_{\hat{s}}$ is increasing in $\hat{s}$. The one-step profit gain shrinks as the gap widens, however, so the return to further innovation declines as a monopoly matures. In an oligopoly, the leader invests in innovation to widen the technology gap and capture a larger profit share, while followers invest to narrow it. Once an industry enters the oligopoly stage, additional entry reduces profits for all incumbents, making mature, crowded industries progressively less attractive to potential entrants. The third part connects prices to industry maturation and to birth cohorts. The wage-adjusted price of a monopoly is pinned down by the fringe technology at the industry's birth. It stays constant during the monopoly phase, and later-born industries start at lower prices. Once an industry becomes an oligopoly, a widening gap, follower entry, and follower catch-up all reduce the price. Because every industry earns the same revenue at each date, falling relative prices translate into rising relative output, matching the industry-level pattern in Stylized Fact 2 of Section **??**.

The static pricing problem therefore maps each industry state into normalized profits and wage-adjusted prices. The profits serve as the flow payoffs in the dynamic problems below. The payoff-relevant states are the gap $\hat{s}$ for a monopoly and the pair (m, n) for an oligopoly. The public knowledge stock s^K determines the technology level at which new industries are born. Innovation and entry move industries across these states. With these payoffs and states in hand, the next subsection characterizes the dynamic decisions of incumbent firms and potential entrants.

4.2 Value Functions and Innovation Decisions

The household's Euler equation,

$$\frac{\dot{C}_t}{C_t} = r_t - \rho,$$

links the interest rate to consumption growth. On a balanced growth path consumption grows at the aggregate rate g, so $r = \rho + g$. After normalizing firm values by $\bar{Y}$, which grows at rate $g - g_N$, this relation produces the effective discount rate $\rho + g_N$ that appears in the Bellman equations below. Here g_N denotes the growth rate of the mass of industries. On the firm side, let $V^M_{\hat{s}}$ denote the value of a monopolist with technology gap $\hat{s}$ over the fringe, and let V_{mn} and V_{-mn} denote the values of a leader and a follower in an oligopoly with technology gap m and n followers.

Because all flow payoffs in a given industry state scale with the common revenue level $\bar{Y}$, it is convenient to express firm values in units of $\bar{Y}$ and work with normalized values $v \equiv V/\bar{Y}$. In a stationary equilibrium these normalized values are time-invariant, so I write $V = v\bar{Y}$ and work directly with v throughout this section (see Appendix B.1).

Monopolist. A monopolist's position is defined by the size of its technology gap over the competitive fringe. This gap determines both how much profit the monopolist can extract today and how much it stands to gain from pushing further ahead. Consider a monopolist with technology gap $\hat{s}$ over the fringe. Its normalized value $v^M_{\hat{s}}$ solves

$$(\rho + g_N)\, v^M_{\hat{s}} = \max_{x\geq 0} \Big\{ \pi^M_{\hat{s}} - R(x) + x\big(v^M_{\hat{s}+1} - v^M_{\hat{s}}\big) + z\,\theta^M\big(v_{01} - v^M_{\hat{s}}\big) - \delta\, v^M_{\hat{s}} \Big\}. \tag{5}$$

The left-hand side is the required flow return on the monopolist's value. It combines time discounting at rate ρ with the dilution of the industry's revenue share as the mass of industries grows at rate g_N. On the right-hand side, the monopolist collects its current normalized profit $\pi^M_{\hat{s}}$ and invests in R&D at cost $R(x)$ to generate innovations at Poisson rate x. Each successful innovation advances the monopolist one step further ahead of the fringe, yielding a capital gain of $v^M_{\hat{s}+1} - v^M_{\hat{s}}$.

The term $z\theta^M(v_{01} - v^M_{\hat{s}})$ captures the threat of entry. θ^M denotes the probability that a potential entrant who meets a monopoly chooses to join rather than create a new industry. This probability is the same for all monopolists, because the new entrant always receives v_{01}, the neck-and-neck duopoly value, regardless of the current $\hat{s}$. At rate $z\theta^M$, a potential entrant chooses to join the monopoly as a follower rather than start a new industry, converting the monopoly into a neck-and-neck duopoly and changing the monopolist's continuation value from $v^M_{\hat{s}}$ to v_{01}. Finally, the monopolist faces exogenous exit at rate δ, destroying its accumulated value.

The optimal innovation intensity is

$$x^M_{\hat{s}} = \left(\frac{\max\{v^M_{\hat{s}+1} - v^M_{\hat{s}},\, 0\}}{\alpha} \right)^{\frac{1}{\psi-1}}.$$

The monopolist's incentive to invest in R&D is governed entirely by the marginal value of an additional step, $v^M_{\hat{s}+1} - v^M_{\hat{s}}$.

Equation (5) and the optimal policy make transparent that the monopolist innovates to widen its technology gap and earn higher profits. Two forces dampen this incentive. First, the marginal gain from pushing an already-large gap further declines, as Lemma 4.1 establishes. Second, the threat of entry limits the horizon over which monopoly rents can be appropriated, since entry converts the industry into an oligopoly and changes the monopolist's continuation value. The decline of $v^M_{\hat{s}+1} - v^M_{\hat{s}}$ with $\hat{s}$ drives the eventual slowdown of monopoly innovation as the industry matures.

Leader and follower. Once a potential entrant joins an incumbent industry, the monopoly transitions into an oligopoly and both the competitive environment and innovation incentives change fundamentally. Within an oligopoly, the leader and followers are locked in a patent race in which each party's payoff depends on the other's actions and on the current state (m, n).

In an industry with a leader m steps ahead of n followers, the leader's normalized value v_{mn} satisfies

$$(\rho + g_N)v_{mn} = \max_{x \geq 0} \Big\{ \pi_{mn} - R(x) + x\big(v_{m+1,n} - v_{mn}\big) \\ + nx_{-mn}\big(v_{m-1,n} - v_{mn}\big) + z\,\theta_{mn}\big(v_{m,n+1} - v_{mn}\big) - \delta v_{mn} \Big\}. \qquad (6)$$

The leader earns flow profits π_{mn} and chooses its innovation intensity x. A successful leader innovation arrives at the chosen rate and widens the gap to $m+1$, raising continuation value by $v_{m+1,n} - v_{mn}$. Followers innovate at common intensity x_{-mn}, so the leader faces a total catch-up intensity nx_{-mn}. A follower innovation narrows the gap to $m-1$ and reduces the leader's continuation value by $v_{mn} - v_{m-1,n}$. In addition, potential entrants arrive at rate z and join this incumbent industry with probability θ_{mn}, which increases the number of followers from n to $n+1$ and changes the leader's continuation value from v_{mn} to $v_{m,n+1}$. As before, exogenous exit arrives at rate δ.

Given follower behavior, the optimal innovation intensity of the leader is

$$x_{mn} = \left(\frac{\max\{v_{m+1,n} - v_{mn},\, 0\}}{\alpha} \right)^{\frac{1}{\psi - 1}}.$$

Equation (6) and the optimal policy highlight the leader's incentive to widen the technology gap. By pulling further ahead, the leader raises its profit share and reduces the threat of follower catch-up. Two channels temper this incentive as the industry matures. As the gap m widens, the marginal value of another step falls, because the leader already captures most of the available rents and followers are too far behind to threaten its position. As the number of followers n rises, the leader's profit share declines and the marginal value of innovation falls with it. The leader's incentive to innovate is therefore strongest in early-stage oligopolies with a small gap and few followers, and weakens as the industry matures into a wide-gap duopoly or becomes crowded.

Each follower's value v_{-mn} solves

$$(\rho + g_N)v_{-mn} = \max_{x\geq 0} \Big\{ \pi_{-mn} - R(x) + \big(x + (n-1)x_{-mn}\big)\big(v_{-m+1,n} - v_{-mn}\big) \\ + x_{mn}\big(v_{-m-1,n} - v_{-mn}\big) + z\,\theta_{mn}\big(v_{-m,n+1} - v_{-mn}\big) - \delta v_{-mn} \Big\}. \quad (7)$$

A follower earns flow profits π_{-mn} and chooses an innovation intensity x. Because follower innovations are shared, the follower benefits not only from its own effort but also from other followers' effort. With total intensity $x + (n-1)x_{-mn}$, the follower group innovates and collectively moves one step closer to the leader, increasing the follower's continuation value from v_{-mn} to $v_{-(m-1),n}$. By contrast, leader innovation arrives at rate x_{mn} and moves the leader further ahead, reducing the follower's continuation value from v_{-mn} to $v_{-(m+1),n}$. Entry at rate $z\theta_{mn}$ adds a follower, tightening competition among followers, and exogenous exit arrives at rate δ.

The common innovation intensity of followers is

$$x_{-mn} = \left(\frac{\max\{v_{-m+1,n} - v_{-mn},\, 0\}}{\alpha} \right)^{\frac{1}{\psi-1}}.$$

The follower's R&D problem highlights a tension between free-riding and catch-up motives. Because improvements diffuse instantaneously among followers, each follower receives the same one-step advance whether it or another follower innovates, while bearing R&D costs only for its own effort. The private return to investing in catch-up nonetheless declines when the leader is already far ahead and when the follower tier is crowded.

When $m = 0$, firms are symmetric neck-and-neck competitors and the value of a representative firm v_{0n} satisfies

$$(\rho + g_N)v_{0n} = \max_{x\geq 0} \Big\{ \pi_{0n} - R(x) + x\big(v_{1n} - v_{0n}\big) \\ + n x_{0n}\big(v_{-1,n} - v_{0n}\big) + z\,\theta_{0n}\big(v_{0,n+1} - v_{0n}\big) - \delta v_{0n} \Big\}. \quad (8)$$

A firm's innovation at rate x moves it into a temporary leader position (state $m = 1$), while innovations by any of the n firms at total rate nx_{0n} can displace its current position. Entry adds a competitor at rate $z\theta_{0n}$, further intensifying business stealing.

This case isolates the escape-competition mechanism familiar from step-by-step models. Competition can raise innovation when firms are close to the frontier, because the prize from becoming the temporary leader is high. At the same time, the model disciplines this force through endogenous entry. As n rises, business stealing intensifies and the net private return to R&D can fall.

The structure of these Bellman equations is standard in step-by-step innovation models. Leaders benefit from pulling ahead and suffer when followers catch up, while followers benefit from their own and their peers' innovation and lose when the leader moves further ahead. Entry shifts the number of followers, exit truncates the firm's horizon, and the Poisson innovation structure ensures that all continuation values are linear in the normalized scale $\bar{Y}$.

Potential entrant. The entry decision is the central margin that links within-industry competition to the rate of industry creation. Recall from Section 3.3 that the cost of creating a new industry, κ, is drawn from a distribution G with support $[\underline{\kappa}, \bar{\kappa}]$, with $0 \leq \underline{\kappa} < \bar{\kappa}$. I restrict attention to equilibria in which $\bar{\kappa} < v_1^M$, so that even the highest-cost potential entrant obtains strictly positive surplus from starting a new industry.

A potential entrant with cost draw κ compares two options. Joining an existing industry is cheap but competitive. The entrant adopts the lowest available technology and earns a follower value, either v_{01} if the industry is currently a monopoly or $v_{-m,n+1}$ if it is an oligopoly with gap m and n followers. Creating a new industry is costly but secures a temporary monopoly. The entrant pays $\kappa\bar{Y}$ and earns v_1^M, the value of a monopolist starting one step above the public knowledge stock. The entrant joins the incumbent industry if and only if the follower value exceeds the net value of creating a new industry, that is, if $v_{S+}^f \geq v_1^M - \kappa$, where v_{S+}^f denotes the entrant's follower value after joining an industry in state S, namely v_{01} if S is a monopoly and $v_{-m,n+1}$ if $S = (m, n)$. The relative attractiveness of the two options depends critically on the state of the industry the entrant encounters. In young, competitive industries where the follower's catch-up rate is high, the follower value v_{S+}^f is high and joining is more attractive. In mature industries where technology gaps are wide and profits are compressed by crowding, follower values are depressed and more entrants find it worthwhile to pay the market initiation cost and create a new industry instead.

These choices can be represented by indicator functions

$$\iota_\kappa^M = \begin{cases} 1 & \text{if } v_{01} \geq v_1^M - \kappa, \\ 0 & \text{otherwise,} \end{cases} \qquad \iota_{mn,\kappa} = \begin{cases} 1 & \text{if } v_{-m,n+1} \geq v_1^M - \kappa, \\ 0 & \text{otherwise.} \end{cases}$$

Aggregating over κ using its distribution G yields the joining probabilities

$$\theta^M \equiv \int \iota_\kappa^M \, dG(\kappa), \qquad \theta_{mn} \equiv \int \iota_{mn,\kappa} \, dG(\kappa).$$

These are the probabilities that a potential entrant who meets a monopoly or an (m, n)-industry decides to join rather than create a new industry. They appear in the Bellman equations (5)-(8), closing the link between the entry decision and incumbent innovation. A higher joining probability in a given state means more frequent follower entry, which lowers leader values and dampens R&D investment by incumbents, while simultaneously reducing the flow of new-industry creation.

Whenever a potential entrant meets an incumbent industry in state S, it creates a new industry with probability $1-\theta(S)$ and joins with probability $\theta(S)$. Since such meetings occur at rate z per incumbent industry, define the gross rate of variety creation per incumbent industry as

$$\gamma \equiv z \, \mathbb{E}[1 - \theta(S)],$$

where the expectation is taken over the state of the industry encountered by the entrant, which coincides with the cross-sectional distribution in the stationary equilibrium characterized below. In a stationary equilibrium, the mass of industries grows at rate $g_N = \gamma - \delta$. Together with ρ, the rate γ determines the effective discount rate $\rho + \gamma$ for normalized firm values.

The next lemma summarizes key properties of the value functions and innovation rates in a stationary equilibrium.

Lemma 4.2 (Properties of value functions and innovation rates). *In any stationary Markov-perfect equilibrium, the normalized value functions and innovation rates satisfy:*

(i) For any normalized value v,

$$0 < v < \frac{1}{\rho + \gamma}.$$

(ii) The monopolist's value $v_{\hat{s}}^M$ increases in its technology gap $\hat{s}$, with

$$\lim_{\hat{s} \to \infty} v_{\hat{s}}^M = \frac{1 + z\theta^M v_{01}}{\rho + \gamma + z\theta^M}.$$

The associated innovation rate $x_{\hat{s}}^M$ is strictly positive and decreases with $\hat{s}$, and satisfies $\lim_{\hat{s} \to \infty} x_{\hat{s}}^M = 0$.

(iii) For any $m \geq 0$, the values v_{mn}, v_{-mn} and their corresponding innovation rates x_{mn}, x_{-mn} converge to 0 as $n \to \infty$. For fixed n,

$$\lim_{m\to\infty} v_{mn} = \frac{1}{\rho+\gamma}, \qquad \lim_{m\to\infty} v_{-mn} = \lim_{m\to\infty} x_{mn} = \lim_{m\to\infty} x_{-mn} = 0.$$

(iv) There exists a finite upper bound $n^* \geq 1$ on the number of followers such that entry into an industry with $n \geq n^*$ followers is never profitable. In particular, $\theta_{mn} = 0$ for all m and all $n \geq n^*$.

Lemma 4.2 characterizes how market structure shapes innovation incentives across different stages of the industry life cycle. Part (i) establishes that normalized firm values are finite and bounded above by $\frac{1}{\rho+\gamma}$, the present value of a perpetual unit profit stream discounted at the effective rate $\rho+\gamma$. This effective rate combines the household's time preference ρ, the dilution of each industry's revenue share as the mass of industries grows at rate g_N, and the exit hazard δ. No firm can be worth more than this bound because flow profits are always below one.

Part (ii) describes how a monopolist's value and innovation effort evolve as it pulls further ahead of the fringe. A larger gap raises current profits, since $\pi^M_{\hat{s}}$ increases in $\hat{s}$, and therefore raises the monopolist's value. However, the marginal profit gain from one additional step diminishes. The incremental return to innovation, $v^M_{\hat{s}+1} - v^M_{\hat{s}}$, therefore shrinks with $\hat{s}$, and the monopolist's R&D effort declines even as its value rises. In the limit, as the gap grows arbitrarily large, its value converges to $\frac{1+z\theta^M v_{01}}{\rho+\gamma+z\theta^M}$. At this limit, the marginal return to further innovation is zero while the entry threat still exists. This declining innovation effort of an established monopolist contributes to the growth slowdown within maturing industries.

Part (iii) reveals how competition shapes innovation incentives within oligopolies along two dimensions, the technology gap m and the number of followers n. As the number of followers grows holding m fixed, profits are competed away by crowding, follower values collapse to zero, and both leaders and followers cease to innovate. Too many competitors chasing the same rents destroy the private return to R&D for all parties. This is the crowding effect. Along the other dimension, as the technology gap grows holding n fixed, the industry converges to a different limit. The leader's value approaches the upper bound $\frac{1}{\rho+\gamma}$, reflecting the near-monopoly profits of a dominant firm with an insurmountable gap, while follower values and innovation rates collapse to zero. Followers in such industries face an increasingly futile catch-up problem. The expected gain from a successful innovation, $v_{-m+1,n} - v_{-mn}$, shrinks as the gap widens, because closing one step still leaves the follower far behind. The leader, meanwhile, faces a negligible competitive threat and has little incentive to invest further. Wide-gap oligopolies thus converge to a low-innovation equilibrium that resembles a de facto monopoly, even though multiple firms remain active.

Part (iv) connects the crowding effect to the entry margin. Follower values fall to zero as n grows, so there exists a finite threshold n^* beyond which no potential entrant finds it worthwhile to join as an additional follower, even at the lowest possible market initiation cost $\underline{\kappa}$. Beyond n^*, all potential entrants who encounter that industry choose to create new industries instead, redirecting the flow of entrepreneurial activity toward the extensive margin, which keeps the relevant set of industry sizes finite in equilibrium.

4.3 Stationary Markov-Perfect Equilibrium

Given the optimal decisions described in the previous subsection, I restrict attention to stationary symmetric Markov-perfect equilibria and characterize their properties. Let $\mu^M_{\hat{s}}$ denote the share of monopoly industries whose leader is $\hat{s}$ steps ahead of the fringe, and let μ_{mn} denote the share of oligopoly industries with technology gap m and n followers. These shares satisfy

$$\sum_{\hat{s}=1}^{\infty} \mu^M_{\hat{s}} \;+\; \sum_{m=0}^{\infty}\sum_{n=1}^{n^*} \mu_{mn} \;=\; 1.$$

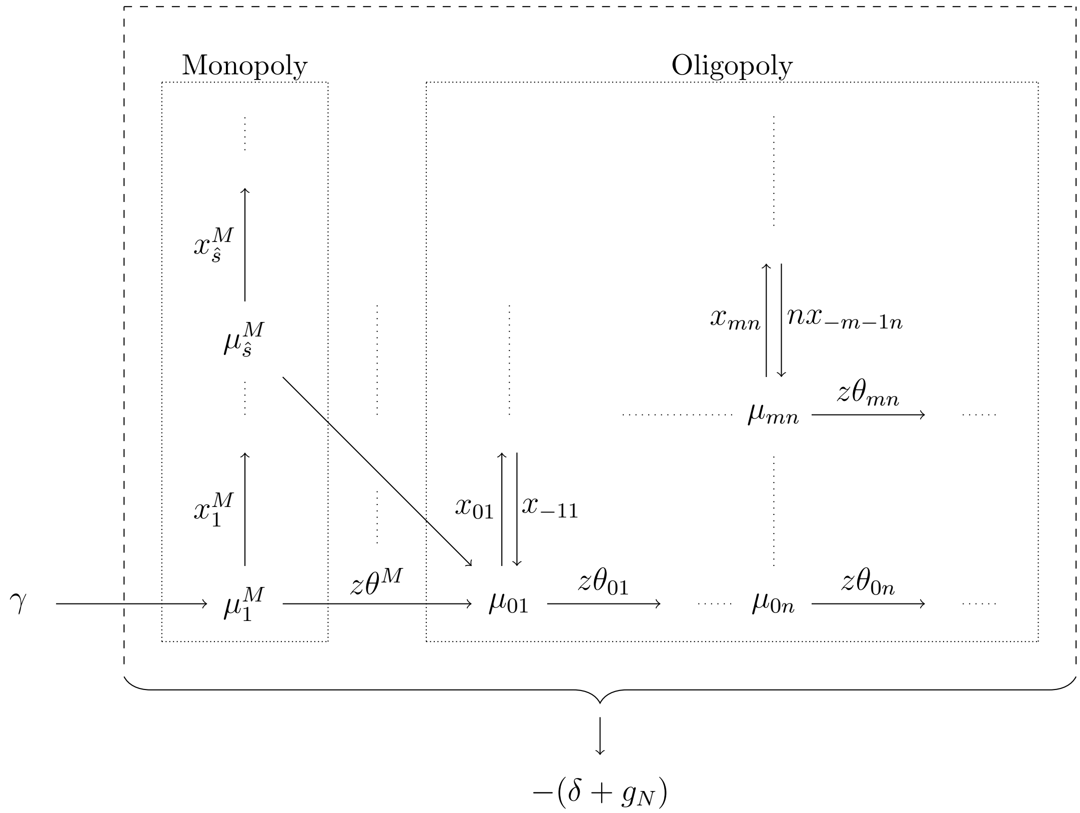
Let $\mu^M = \sum_{\hat{s}=1}^{\infty} \mu^M_{\hat{s}}$ denote the total share of monopolies. The gross rate of variety creation γ, defined in Section 4.2 as $z\,\mathbb{E}[1-\theta(S)]$, can now be expressed explicitly using the cross-sectional distribution:

$$\gamma \;=\; z\left[(1-\theta^M)\mu^M + \sum_{n=1}^{n^*}\sum_{m=0}^{\infty}(1-\theta_{mn})\mu_{mn}\right].$$

Here, z is the exogenous arrival rate of potential entrants, while γ is the endogenous equilibrium rate of new-industry creation. This expression formalizes the relationship between entry decisions and aggregate growth that was previewed in Section 4.2. The rate γ aggregates the fraction of potential entrants who choose to create new industries rather than join existing ones, weighted by the mass of industries in each state. A higher fraction of mature industries, where follower values are depressed and joining is unattractive, diverts more entrepreneurial activity toward industry creation and raises γ. The Kolmogorov forward equations that determine $\{\mu^M_{\hat{s}}, \mu_{mn}\}$ in a stationary equilibrium are given in the appendix.

To understand what shapes the stationary distribution, it is useful to trace the life cycle of a typical industry through the state space. Figure 1 summarizes the resulting flows. Every industry enters the economy as a fresh monopoly at $\hat{s} = 1$, contributing to the flow γ of newly created industries. From there, monopoly industries evolve along two margins. First, the monopolist climbs the quality ladder through successful innovation. At rate $x^M_{\hat{s}}$,

Figure 1: Flowchart of industry shares



the monopolist advances from $\hat{s}$ to $\hat{s}+1$, pulling further ahead of the fringe and raising its profit. Second, at rate $z\theta^M$, a potential entrant chooses to join the monopoly as a follower rather than create a new market. This converts the monopoly into a neck-and-neck duopoly at state $(0, 1)$ and initiates the oligopoly phase of the life cycle.

Once an industry enters the oligopoly phase, its evolution is governed by the interplay of leader and follower innovation and continued entry. The technology gap widens at rate x_{mn} when the leader innovates and narrows at the total rate nx_{-mn} when any of the n followers successfully catches up. At the same time, new followers join at rate $z\theta_{mn}$, pushing the industry toward more crowded states. As Lemma 4.2 establishes, this crowding progressively depresses both leader and follower values and weakens innovation incentives for all parties. Joining probabilities eventually fall to zero once the industry reaches n^* followers. An industry therefore traverses a predictable arc. It is born competitive, passes through an oligopolistic phase of active innovation and entry, and gradually settles into a mature, low-innovation configuration before exiting at rate δ.

The total mass of industries N grows at the net rate

$$g_N = \gamma - \delta.$$

A surviving industry therefore loses relative share at rate g_N as new industries enter, in addition to facing exit at rate δ. In a stationary equilibrium, the distribution $\{\mu^M_{\hat{s}}, \mu_{mn}\}$ is time-invariant, meaning that the flows of industries into and out of each state exactly balance. The key aggregate state variables for growth are therefore the rate of variety creation γ and the distribution of gaps and followers summarized by $\{\mu^M_{\hat{s}}, \mu_{mn}\}$. I can now define the equilibrium formally.

Definition 4.3 (Equilibrium). *A stationary and symmetric Markov-perfect equilibrium consists of values $\{v^M_{\hat{s}}, v_{mn}, v_{-mn}\}$, policies $\{x^M_{\hat{s}}, x_{mn}, x_{-mn}, \theta^M, \theta_{mn}\}$, an industry distribution $\{\mu^M_{\hat{s}}\}_{\hat{s}} \cup \{\mu_{mn}\}_{m,n}$, and scalars $\{g_N, \gamma\}$ such that all individual decisions solve the corresponding optimization problems and the laws of motion for industry shares are satisfied.*

With the equilibrium distribution in hand, I can characterize how the aggregate technology level evolves. Let $\bar{s}^M_t$ denote the average monopolist technology level and $\bar{s}^{cf}_t$ the average technology level of the associated competitive fringe firms. For oligopoly industries, let $\bar{s}^f_t$ denote the average lowest technology level and $\bar{s}^l_t$ the average highest technology level, and define the average technology gap among oligopolies as $\bar{m}$. The public knowledge stock can then be written as

$$s^K_t = \mu^M \bar{s}^M_t + (1 - \mu^M)\bar{s}^f_t.$$

Lemma 4.4 (Properties of technology levels in a stationary equilibrium). *The public knowledge stock grows at a constant rate*

$$\dot{s}^K = \gamma + \sum_{\hat{s}=1}^{\infty} \mu^M_{\hat{s}} x^M_{\hat{s}} + \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} n x_{-mn} \mu_{mn}.$$

The average technology levels satisfy

$$\begin{aligned}
\gamma(\bar{s}^M - \bar{s}^{cf}) &= \gamma + \sum_{\hat{s}=1}^{\infty} \mu^M_{\hat{s}} x^M_{\hat{s}}, \\
\gamma(1-\mu^M)(\bar{s}^l - \bar{s}^f) &= \gamma(1-\mu^M)\bar{m} \\
&= \sum_{n=1}^{n^*} \left[(n+1) x_{0n} \mu_{0n} + \sum_{m=1}^{\infty} x_{mn} \mu_{mn} \right] - \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} n x_{-mn} \mu_{mn}.
\end{aligned}$$

Lemma 4.4 characterizes the technology dynamics that underpin steady aggregate growth. The first result establishes that the public knowledge stock grows at a constant rate in a stationary equilibrium, combining three distinct sources of productivity advance. The first is variety creation at rate γ. Each new industry is born at a productivity level one step above the current public knowledge stock, so the creation of new industries contributes to the growth of s^K even before any within-industry innovation occurs. The second and third terms capture within-industry innovation by monopolists and followers respectively. Monopolists raise the public knowledge stock by innovating ahead of the fringe, since the monopolist is the sole active producer and its technology sets the industry's lowest active tier. Followers contribute by catching up to the leader, which advances the lowest technology tier in oligopolies and thereby also raises the public knowledge stock.

The second set of expressions in Lemma 4.4 links the average technology gaps to the balance of innovative activity across sectors. The average lead of monopolists over the fringe, $\bar{s}^M - \bar{s}^{cf}$, is maintained in steady state by opposing forces. New monopoly creation and monopolist innovation widen the average gap, while the expansion of the industry mass at rate γ continuously dilutes it. Similarly, the average oligopoly gap $\bar{m}$ is kept constant by the balance between leader innovation, which widens gaps, follower innovation, which narrows them, and the dilution by industry mass growth. These stationarity conditions are not mechanical accounting identities but reflect equilibrium restrictions. They require that the cross-sectional distribution of industries across states be consistent with the individual innovation incentives characterized in Section 4.2. Because these shares are equilibrium objects rather than free parameters, the same distribution that determines $\dot{s}^K$ here reappears as the weights in the aggregate growth decomposition of Section 5, which I turn to next.

5 Aggregate Growth and Industry Dynamics

This section derives the main theoretical results. I first decompose the aggregate growth rate into contributions from variety creation, monopolist innovation, and catch-up innovation, each weighted by the stationary distribution characterized in Section 4. I then characterize how individual industries evolve over their life-cycles despite the stationarity of the cross-section. Finally, I analyze the constrained efficiency of the industry creation margin and identify the externalities that drive a wedge between the private and social incentives to create new industries.

5.1 Aggregate Growth Decomposition

Section 4 characterized the stationary cross-sectional distribution of industries, with constant shares in each life-cycle state and a well-defined rate of variety creation γ. I now show how the aggregate growth rate emerges from this stationary distribution. The central result is that steady aggregate growth arises even though individual industries follow non-stationary life-cycle paths. The stationary cross-section ensures that the average contribution of each growth margin remains constant over time.

Proposition 5.1 (Steady growth with non-stationary industry life-cycle). *In any symmetric stationary Markov-perfect equilibrium, the constant growth rate can be expressed as:*

$$
\begin{aligned}
g &= \ln\lambda\left(\gamma + \sum_{\hat{s}=1}^{\infty}\mu_{\hat{s}}^{M}x_{\hat{s}}^{M} + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}\right)\\
&= \ln\lambda\left(\gamma + \sum_{\hat{s}=1}^{\infty}\mu_{\hat{s}}^{M}x_{\hat{s}}^{M} + \sum_{n=1}^{n^*}\left((n+1)x_{0n}\mu_0 + \sum_{m=1}^{\infty} x_{mn}\mu_{mn}\right) - \gamma(1-\mu_M)\bar{m}\right)\\
&= \ln\lambda\cdot \dot{s}^K.
\end{aligned}
$$

The first line decomposes the growth rate into three distinct sources, each scaled by the step size $\ln\lambda$. The first term, γ, captures the contribution of variety creation. Each new industry is born with a productivity level one step above the current public knowledge frontier, so the flow of new industries directly raises the average technology level. A new monopolist enters at rate γ with a lowest active tier of s^K+1. Incumbent industries lose share over the same interval at the same rate $\gamma=\delta+g_N$. The lowest tier of a departing industry averages s^K, so the s^K levels cancel between entry and exit. Each new industry then contributes one quality step, for a net flow of γ. The second term aggregates the innovation effort of monopolists weighted by their share $\mu_{\hat{s}}^{M}$, capturing how incumbents push the technology frontier upward within existing monopoly industries. The third term captures catch-up innovation in oligopolies where followers innovate at the combined rate nx_{-mn}, advancing the lowest technology tier and thereby raising the public knowledge stock even without moving the frontier.

The second line rewrites the same expression in terms of the evolution of leaders' technology levels and the stationary gap $\bar{m}$, using Lemma 4.4. This alternative decomposition replaces the catch-up innovation terms with leader innovation, net of the correction term $\gamma(1-\mu_M)\bar{m}$. This correction arises because new industries replace exiting ones starting at s^K+1. Only the s^K component of the exiting industry's technology is replaced by the first γ term. The above-lowest-level component of oligopoly leaders, which on average equals $\bar{m}$, disappears upon exit and is not replenished by the new entrant. The last equality simply notes that the sum of these contributions is, by definition, the rate of growth of the public knowledge stock $\dot{s}^K$.

Proposition 5.1 shows that aggregate growth is fully determined by industry dynamics. Variety creation, innovation in monopoly industries, and catch-up innovation in oligopolies all contribute to growth, and their relative importance depends on the stationary distribution of industries across life-cycle states. This structure implies that policies or parameter changes that shift the cross-sectional distribution affect aggregate growth through several margins simultaneously. Making entry into existing industries more or less attractive, for instance, changes the balance of all three growth components. I return to this point in Section 5.3.

The existence of a stationary equilibrium relies on two features of the model working in tandem. Private innovation incentives are disciplined by diminishing marginal gains from pushing an already-large lead further and by intensifying competition from an expanding follower base, which jointly make innovation and profits negligible in very mature, crowded industries. Endogenous industry creation then continually replenishes the cross-section with new, innovation-intensive monopolies, offsetting the gradual maturation and exit of incumbent industries. Without the first feature, the cross-sectional distribution would unravel as gaps grow without bound. Without the second, the economy would eventually be populated only by mature, low-innovation industries and growth would stall.

5.2 Industry Life-Cycle Dynamics

The entry decision that determines aggregate growth also governs how individual industries evolve over their life-cycles. The stationary cross-sectional distribution characterized in Section 4.3 is not sustained because industries individually reach a steady state. Rather, the maturation of incumbent industries is continuously offset by the creation of new ones and the exit of old ones. The following results characterize the implied dynamics of surviving industries and connect them to the empirical patterns documented in Section 2.

Proposition 5.1 characterized steady aggregate growth using stationary cross-sectional objects. A stationary cross-sectional distribution does not imply that old surviving industries are representative of the cross-section. Because industries are continually replenished by newborn oligopolies, selection makes surviving cohorts older and more mature on average. Cohort-level technology gaps can therefore drift upward over the life-cycle even while the cross-sectional mean gap remains constant.

Let $(\tilde{m}, \tilde{n})$ denote an oligopoly state drawn from the stationary cross-sectional distribution μ conditional on being an oligopoly, and let $\tilde{m}_\Delta$ denote the technology gap after a short interval Δ under the within-oligopoly transition dynamics, starting from the cross-sectional draw $(\tilde{m}, \tilde{n})$. Let $\hat{m}_a$ denote the realized technology gap of an oligopoly industry at age a, where age is measured from the first time the industry has at least one follower, so that $a = 0$ corresponds to a newborn oligopoly. Define

$$f(a) \equiv \mathbb{E}[\hat{m}_a \mid \text{alive at age } a] \,.$$

Let $U \sim \text{Exp}(\gamma)$ denote the age of a randomly sampled surviving oligopoly from the stationary cross-section. Then

$$\bar{m} = \mathbb{E}[f(U)], \qquad \bar{m}_\varepsilon = \mathbb{E}[f(U) \mid U > \varepsilon].$$

That is, $\bar{m}_\varepsilon$ is the average technology gap among surviving oligopolies older than ε.

Assumption 5.1. *For sufficiently mature surviving oligopolies, leader innovation weakly dominates catch-up innovation in expectation, so the expected instantaneous change in the technology gap is weakly positive.*

Proposition 5.2 (Cross-sectional and cohort technology-gap dynamics)**.** *In a stationary Markov-perfect equilibrium with stationary cross-sectional industry distribution μ, oligopoly technology gaps satisfy:*

(i) In the stationary cross-section, technology gaps tend to widen on average at the industry level:

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}\big[\tilde{m}_\Delta - \tilde{m}\big] = \gamma \bar{m} > 0.$$

(ii) Older surviving oligopolies have above-average gaps: for sufficiently small $\varepsilon > 0$,

$$\bar{m}_\varepsilon > \bar{m}.$$

(iii) Under Assumption 5.1, $f(a)$ is weakly increasing for sufficiently large ages.

Proposition 5.2 makes clear that stationarity in this model is an aggregate and cross-sectional property, not an industry-level one. Part (i) shows that even when an oligopoly is drawn from the stationary cross-section, its technology gap tends to widen on average under the within-industry dynamics. Stationarity of the cross-sectional mean pins down this drift. Newly formed oligopolies enter at gap zero and dilute the cross-section at rate γ, so the widening of surviving oligopolies must equal $\gamma \bar{m}$ to hold the mean gap constant.

Part (ii) is the cohort counterpart of this observation. The stationary cross-section places disproportionate weight on young oligopolies, which begin with low technology gaps. As a result, once a sufficiently small neighborhood of newborn surviving oligopolies is excluded, the average technology gap exceeds the stationary cross-sectional mean. A stationary cross-section is therefore not representative of surviving cohorts. This result does not yet imply a general monotone age profile, however, which requires the additional condition in Part (iii).

Part (iii) strengthens the result to a monotone age profile under the additional condition of Assumption 5.1 on the balance of leader and follower innovation in mature industries. This stronger cohort result again relies on endogenous industry creation. New industries are continually born with low technology gaps, which replenishes the cross-section with young oligopolies even as surviving cohorts become progressively more mature. In the calibrated economy, I verify numerically that Assumption 5.1 is satisfied and that $f(a)$ is strictly increasing over the relevant range of ages. As technology gaps widen along surviving cohorts, the marginal return to innovation falls for both leaders and followers, entry becomes less attractive, and leadership turnover becomes rare.

Since follower counts do not decrease in the baseline model, Proposition 5.2 implies that older surviving oligopolies tend to have both larger technology gaps and weakly larger follower bases. Larger gaps and more followers lower relative prices and raise industry quantities. The next proposition collects these life-cycle implications and connects them to the empirical regularities documented in Section 2.

To state the life-cycle implications, define the following industry-level variables. Let $P_{mn}(s^f)$ denote the industry price of an oligopoly with technology gap m, n followers, and follower technology level s^f. Let $Y_{mn}(s^f)$ denote the corresponding industry output.

Proposition 5.3 (Properties of industry life-cycle)**.** *In a stationary Markov perfect equilibrium, conditional on industry survival:*

(i) Monopoly quality growth slows:

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}\big[\ln q^M_{a+\Delta} - \ln q^M_a \,\big|\, \text{still monopoly, alive}\big] \searrow 0 \qquad \text{as } a \to \infty.$$

(ii) Mature oligopoly states have lower relative prices and higher quantities: For each $m \geq 0$, $n \in \{1, \ldots, n^* - 1\}$, and s_f,

$$\frac{P_{m+1,n}(s_f)}{w} \leq \frac{P_{mn}(s_f)}{w}, \qquad Y_{m+1,n}(s_f) \geq Y_{mn}(s_f),$$

and

$$\frac{P_{m,n+1}(s_f)}{w} \leq \frac{P_{mn}(s_f)}{w}, \qquad Y_{m,n+1}(s_f) \geq Y_{mn}(s_f).$$

(iii) Under Assumption 5.1, for sufficiently large ages, surviving oligopoly cohorts move toward lower-price, higher-quantity, lower-innovation, lower-entry, and lower-turnover states.

Proposition 5.3 summarizes the industry-level forces behind the model's life-cycle patterns.

Part (i) characterizes the monopoly stage. As a surviving monopolist extends its lead over the fringe, the marginal gain from further innovation declines. Monopoly quality growth therefore slows over time until the industry eventually attracts entry and transitions into oligopoly.

Part (ii) characterizes oligopoly states. Holding follower technology and the number of followers fixed, a larger technology gap lowers the industry price relative to the wage and raises industry quantity. Holding the technology gap and follower technology fixed, a larger follower base has the same effect through stronger within-industry competition. These state-level results imply that, as surviving oligopoly cohorts move toward larger gaps and weakly larger follower counts, mature industries tend to become lower-price and higher-quantity in the within-industry sense. This provides a partial micro-foundation for the life-cycle evidence on falling prices and rising industry output.

Part (iii) translates the cohort gap dynamics of Proposition 5.2 into broader life-cycle implications. As surviving oligopoly cohorts age, they move toward lower-price, higher-quantity, lower-innovation, lower-entry, and lower-turnover states. The industry life-cycle evidence is consistent with this pattern. Early phases are marked by rapid innovation and entry, while later phases are characterized by consolidation and lower technological activity (Abernathy and Utterback, 1978; Gort and Klepper, 1982; Klepper, 1996; Filson, 2001; Jovanovic and Wang, 2025).

The economy therefore does not converge to a representative mature industry. It is precisely the entry of new, innovation-intensive industries that prevents the cross-section from being dominated by mature, low-growth sectors. I now turn from these positive results to the efficiency of the industry creation margin.

5.3 Constrained efficiency of industry creation

Proposition 5.1 showed that aggregate growth depends on the stationary distribution of industries, which is shaped by entrants' choices between joining incumbent industries and creating new ones. A natural question is whether the equilibrium produces the right mix of industry creation and incumbent entry. To address this, I compare the equilibrium allocation with a constrained planner's benchmark.

The equilibrium entry decision in Section 4.2 is formulated as a binary choice for each potential entrant with a realized cost draw κ, between joining the encountered incumbent industry and creating a new one. To compare the equilibrium allocation with a social benchmark, it is useful to rewrite this decision in terms of state-specific cost thresholds. Let S index an industry state, where $S = M$ denotes a monopoly with technology gap $\hat{s}$ and $S = mn$ denotes an oligopoly with gap m and n followers. Let μ_S be the stationary share of industries in state S and let v^f_{S+} denote the follower value after joining that industry. For example, $v^f_{S+} = v_{01}$ if the industry is a monopoly and $v^f_{S+} = v_{-m,n+1}$ if it is an oligopoly with gap m and n followers.

Before being randomly assigned to an incumbent industry and before observing its own cost draw κ, a potential entrant chooses a vector of thresholds $\{\hat{\kappa}^{CE}_S\}_S$ that solves

$$\max_{\{\hat{\kappa}_S\}_S} \sum_S \mu_S \left[\int_{\underline{\kappa}}^{\hat{\kappa}_S} \left(v^M_1 - \kappa\right) dG(\kappa) + \int_{\hat{\kappa}_S}^{\overline{\kappa}} v^f_{S+}\, dG(\kappa) \right], \tag{9}$$

where G is the distribution of market initiation costs and v^M_1 is the value of a new monopolist.[3] In each state S, the entrant draws a cost κ and compares two options. If $\kappa < \hat{\kappa}_S$, the entrant pays κ to create a new industry and earns $v^M_1 - \kappa$. If $\kappa \geq \hat{\kappa}_S$, the entrant joins the incumbent industry and earns the follower value v^f_{S+}. The first-order condition for an interior threshold in state S is

$$\mu_S G'(\hat{\kappa}^{CE}_S) \left[-\hat{\kappa}^{CE}_S + v^M_1 - v^f_{S+} \right] = 0, \tag{10}$$

so that whenever $\mu_S G'(\hat{\kappa}^{CE}_S) > 0$ the equilibrium cutoff satisfies

$$\hat{\kappa}^{CE}_S = v^M_1 - v^f_{S+} \approx \frac{\pi^M_1 - \pi^f_{S+}}{\rho + \gamma}. \tag{11}$$

The approximation on the right replaces the full value difference with the flow profit differential $\pi^M_1 - \pi^f_{S+}$ capitalized at the effective discount rate $\rho + \gamma$. The implied incumbent joining probability is

$$\theta_S \;=\; 1 - G\big(\hat{\kappa}^{CE}_S\big) \tag{12}$$

and it coincides with the probabilities defined in the entry problem above.

To assess the efficiency of this allocation, I consider a constrained planner who can intervene only in the industry creation margin. The planner does not directly control incumbent pricing or innovation decisions. Instead, the planner chooses a vector of thresholds $\{\hat{\kappa}^{SP}_S\}_S$ to maximize the representative household's lifetime utility, subject to the aggregate resource constraint and the induced law of motion for the industry distribution. Firms continue to optimize given the competitive environment they face. However, the planner recognizes that changing entry thresholds alters the stationary distribution of industries, which in turn affects firm values and innovation incentives through the equilibrium fixed-point.

For any given set of thresholds $\{\hat{\kappa}_S\}_S$, the induced entry probabilities $\{\theta_S\}_S$ determine a unique stationary distribution over industry states and a corresponding balanced growth

[3] I assume $\underline{\kappa} < v^M_1$ so that starting a new industry yields strictly positive surplus at low costs.

path for consumption. Let $\mathcal{R}(\{\hat{\kappa}_S\}_S)$ denote the ratio of incumbent R&D expenditure to aggregate output and $\mathcal{K}(\{\hat{\kappa}_S\}_S)$ the ratio of market initiation costs to output as defined in Appendix B.4, and $g(\{\hat{\kappa}_S\}_S)$ the implied aggregate growth rate. With log preferences, the planner's welfare can be written as

$$W\big(\{\hat{\kappa}_S\}_S\big) = \frac{1}{\rho}\ln\big(1 - \mathcal{R}\big(\{\hat{\kappa}_S\}_S\big) - \mathcal{K}\big(\{\hat{\kappa}_S\}_S\big)\big) + \frac{1}{\rho^2}\, g\big(\{\hat{\kappa}_S\}_S\big), \tag{13}$$

where the first term captures the level of net consumption on the balanced growth path and the second term captures the contribution of long-run growth to lifetime welfare. The coefficient $1/\rho^2$ arises from log preferences and reflects that a permanent increase in the growth rate affects consumption at all future dates. Because these gains accumulate linearly over time and are discounted at rate ρ, a one-unit increase in g raises welfare by $1/\rho^2$.

The constrained planner's first-order condition for the cutoff in state S can be written as

$$\frac{z}{\rho}\,\mu_S G'\big(\hat{\kappa}_S^{SP}\big)\left[-\frac{1}{1-\mathcal{R}-\mathcal{K}}\,\hat{\kappa}_S^{SP} + \frac{1}{\rho}\ln\lambda\right] + \Phi_S = 0, \tag{14}$$

where all objects are evaluated at $\{\hat{\kappa}_S^{SP}\}_S$. The term $\frac{z}{\rho}\mu_S$ is the mass of potential entrants throughout history who encounter an industry in state S, and $G'(\hat{\kappa}_S^{SP})$ weights the marginal entrant at the cutoff. Inside the brackets, $-\frac{1}{1-\mathcal{R}-\mathcal{K}}\hat{\kappa}_S^{SP}$ is the utility cost of the market initiation expenditure, evaluated at the shadow value of consumption. The term $\frac{1}{\rho}\ln\lambda$ is the welfare gain from permanently raising the knowledge level by one quality step. The term Φ_S collects the marginal welfare effect of changing industry composition and incumbent innovation rates when the vector of thresholds is perturbed.

Comparing the planner's first-order condition (14) with the private entrant's optimality condition (10) reveals the sources of inefficiency in the decentralized equilibrium. The private entrant equates the market initiation cost $\hat{\kappa}_S^{CE}$ to the value differential $v_1^M - v_{S+}^f$, which is approximately $\frac{\pi_1^M - \pi_{S+}^f}{\rho+\gamma}$. The planner instead weighs the shadow cost $\frac{1}{1-\mathcal{R}-\mathcal{K}}\hat{\kappa}_S^{SP}$ against the social benefit $\frac{\ln\lambda}{\rho}$, plus the composition adjustment Φ_S. Four distinct forces drive a wedge between these two cutoffs. Three operate through the direct terms inside the brackets of (14), while a fourth arises from the general equilibrium response of the industry distribution captured by Φ_S. I discuss each in turn.

A *resource-cost externality* distorts the cost valuation. Private firms treat the market initiation cost as an expense in output units, whereas the planner applies the shadow value $\frac{1}{1-\mathcal{R}-\mathcal{K}}$ to account for the aggregate resource constraint. Higher aggregate investment in R&D and market creation raises the marginal utility of consumption, making the utility cost of funding industry creation strictly higher than its output-unit cost.

A *knowledge externality* drives a second wedge. Private entrants value industry creation through the flow of appropriable rents, $\pi_1^M - \pi_{S+}^f$, whereas the planner values the permanent expansion of the knowledge stock by one quality step λ. Since $\ln\lambda > \pi_1^M = 1-\lambda^{-1}$ for $\lambda > 1$, the private rent necessarily falls short of the social return to industry creation. Potential entrants undervalue the contribution of new industries to the aggregate knowledge stock because they cannot appropriate the full productivity gain.

A *creative-destruction externality* amplifies this wedge through the discount rate. The private discount rate, $\rho+\gamma$, is strictly larger than the social discount rate, ρ, because new industries arrive at rate γ and erode incumbent market shares. Private firms therefore discount future profits more heavily than is socially warranted. The planner discounts knowledge accumulation only at the time preference rate ρ, recognizing that the productivity gain $\ln\lambda$ persists beyond the life of the firm.

An *equilibrium adjustment effect* completes the comparison. The term Φ_S embeds the general equilibrium response of values, policies, and industry shares when the entire vector of thresholds is perturbed. A local change in $\hat{\kappa}_S$ affects not only the flow of entrants into state S but also the induced evolution of other states through the stationary distribution and the aggregate resource constraint. This adjustment further shifts the planner's preferred cutoff away from the private one. In the next section, I evaluate these conditions in the calibrated economy and quantify the resulting entry wedge.

6 Quantitative Analysis

This section disciplines the theoretical framework quantitatively. I calibrate a stationary equilibrium to post-2000 U.S. moments, using 2000–2019 when available, and then use the calibrated economy to address three questions. First, does the model generate non-stationary industry life-cycles that are qualitatively consistent with the empirical patterns documented in Section 2? Second, how do the composition and intensity of entrepreneurial entry affect aggregate growth? Third, does the decentralized economy create too few new industries, as the constrained efficiency analysis in Section 5.3 predicts, and what does this wedge imply for the design of innovation policy?

Table 1: Calibrated Parameters

Externally calibrated		
ρ	0.0251	Rate of time preference (10-year Treasury rate − TFP growth)
δ	0.0834	Exit rate (exit/entry, (Broda and Weinstein, 2010; Adhami, 2025))
Internally calibrated		
σ	3.4939	Elasticity of substitution
λ	1.0125	Innovation step size
α	0.0363	Multiplier of R&D cost function
ψ	1.7190	Curvature of R&D cost function
$\hat{k}$	0.0505	Mean of market initiation cost divided by $\tilde{v}_1^M$
z	0.1539	Potential entrant arrival rate

6.1 Calibration

For the quantitative analysis, I parameterize the distribution of market initiation costs as follows. Let κ denote the cost of creating a new industry, and write

$$\kappa = k\, \tilde{v}_1^M,$$

where k is drawn from a transformed beta distribution with support $[0, 0.9]$ and $\tilde{v}_1^M$ is the value of a monopolist in the baseline equilibrium.[4] In comparative statics, I vary the mean of this distribution while holding its standard deviation fixed at 0.1.

I calibrate eight parameters to reflect the U.S. economy over the 20-year period starting in 2000. Table 1 reports the parameter values. Two parameters are chosen externally. The rate of time preference ρ is set to 0.0251, the difference between the average 10-year Treasury rate and average TFP growth over 2000–2019, where TFP is measured using the utilization-adjusted series from Fernald (2014). The industry exit rate δ is set to 0.0834, based on exit-to-entry patterns in Broda and Weinstein (2010) and Adhami (2025). [5]

The remaining six parameters, $\{\sigma, \lambda, \alpha, \psi, \hat{k}, z\}$, are estimated jointly to match the empirical moments in Table 2. Appendix C.2 provides the estimation procedure and the construction of each moment. While all parameters simultaneously affect all moments in general equilibrium, the following discussion identifies which data moments primarily pin down each parameter.

The elasticity of substitution σ governs the intensity of static price competition, while

[4]This normalization ties the support of market initiation costs to the value of a new monopolist, which is convenient for comparing cutoffs across equilibria.

[5]This corresponds to an annual exit probability of approximately 0.08 under the conversion $\delta = -\ln(1 - 0.08)$. The exit rate does not affect the variety-creation rate γ, firm values, or entry decisions. It shifts only the level of g_N, so I calibrate it coarsely.

Table 2: Targeted moments

	Data	Model	Source
Average TFP growth	0.0086	0.0086	Fernald (2014) (2000–2019)
Cost-weighted average markup	1.4111	1.4321	Compustat, DLEU (2000–2016)
Cost-weighted S.D. of log markup	0.2640	0.2528	Compustat, DLEU (2000–2016)
Average R&D/GDP	0.0269	0.0267	NCSES (2000–2019)
Average innovation frequency	0.5057	0.5064	DISCERN (2000–2019)
Average brand entry rate	0.0850	0.0861	ALM (2007–2014)

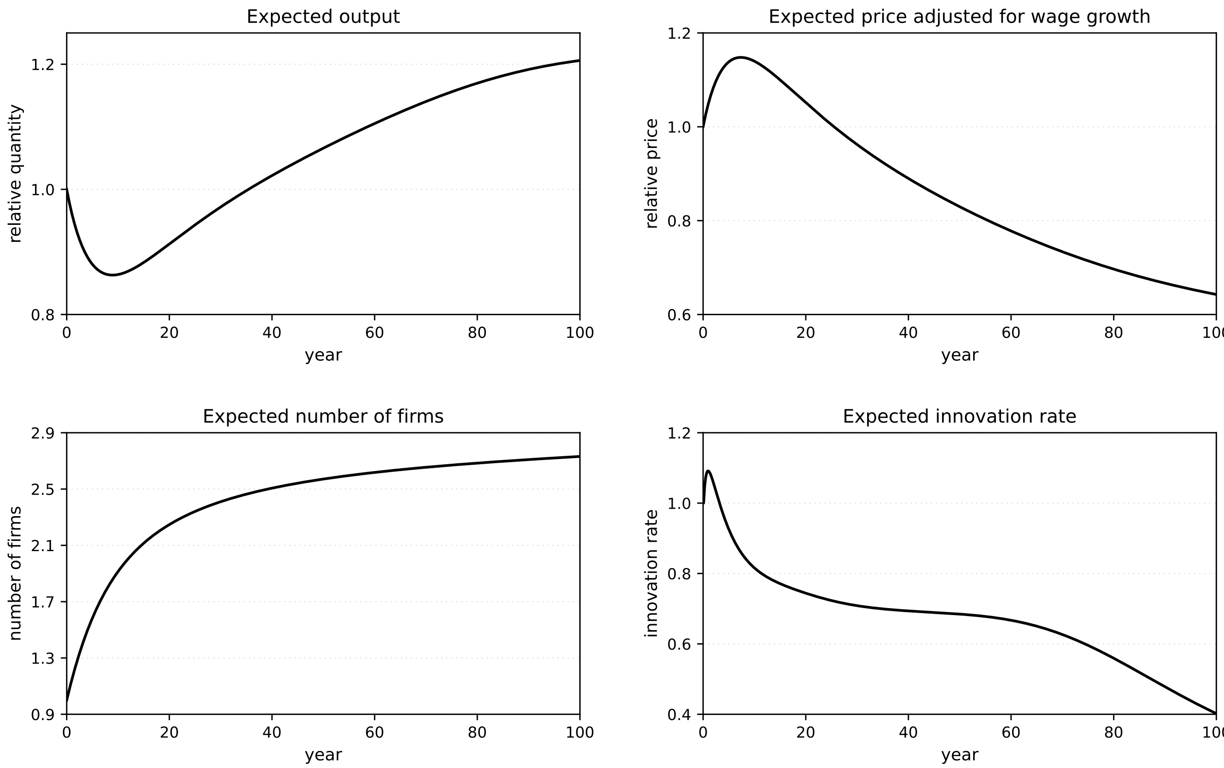
Notes: DLEU denotes De Loecker, Eeckhout and Unger (2020). NCSES denotes National Center for Science and Engineering Statistics (NCSES). DISCERN denotes the firm-level patent panel of Arora, Belenzon and Sheer (2021). ALM denotes Argente, Lee and Moreira (2024). Markup moments are computed from Compustat using the production approach of DLEU. The brand entry target is an annual per-brand entry rate; it is distinct from z, the model's potential-entrant arrival rate per industry.

the market initiation cost parameter $\hat{k}$ governs the extensive margin of entry. Together they determine the level and dispersion of within-industry competition. These parameters are primarily disciplined by the average cost-weighted markup and the cost-weighted dispersion of log markups. I compute markups on Compustat firms using the production approach of De Loecker, Eeckhout and Unger (2020, DLEU), combining their industry-level output elasticities with firm-level sales and costs over 2000–2016. For the dispersion moment, I take the cost-weighted standard deviation of log markups after removing four-digit-industry and year fixed effects, so that the target reflects within-economy heterogeneity across firms rather than permanent cross-industry differences in markup levels or aggregate movements common to all firms. This is the empirical counterpart to the model's cross-sectional markup dispersion. Appendix C.2 reports the full construction.

The R&D cost parameters (α, ψ) govern the cost of innovation for incumbents. I discipline them using the average R&D-to-GDP ratio over 2000–2019, from National Center for Science and Engineering Statistics (NCSES), and the average innovation frequency. I measure innovation frequency in the DISCERN firm-level panel (Arora, Belenzon and Sheer, 2021) over 2000–2019, as the fraction of firm-year observations in that panel in which a firm is granted at least one patent. The model counterpart is the probability that a firm experiences at least one innovation event in a year, $1 - \exp(-h)$, where h is the firm's annual innovation hazard. Appendix C.2 details this mapping. In the model, both incumbent R&D spending and market initiation costs are counted as R&D, since both represent forms of innovation.

The innovation step size λ determines the return to innovation and is the primary driver of the aggregate growth rate. I calibrate it to match the average TFP growth rate. Finally, z governs the frequency of potential entrant shocks. In the model, each potential entrant shock corresponds to the creation of a new firm or brand, either by joining an incumbent industry

Figure 2: Industry Life-Cycle Patterns



or by creating a new industry. I therefore discipline z by matching the model-implied annual per-brand entry rate, $z/\bar{F}$, to the sales-weighted average brand entry rate from Argente, Lee and Moreira (2024, ALM), where $\bar{F}$ is the model-implied average number of firms per industry.

6.2 Industry Life-Cycle Patterns

The defining feature of the model is that endogenous industry creation generates non-stationary life-cycles whose cross-sectional composition determines aggregate growth. In the calibrated economy, industries follow precisely this pattern. They start small and highly innovative, then grow in size while innovation activity gradually slows. The profiles in Figure 2 show that the calibrated economy produces life-cycle dynamics that are qualitatively consistent with the empirical evidence discussed in Section 2. Each panel shows the expected change in an industry variable over a 100-year period, conditional on survival. Details of the calculation are provided in Appendix C.3.

The top-left panel plots model output relative to its initial level. The output measure is the industry-level aggregator in the model, which is best interpreted as the output of a narrowly defined market, close to a group of brands that compete directly. In practice, newly created varieties may compete with incumbent products before appearing as separate industry classifications. Standard industry-level measures can therefore partly absorb business stealing across nearby varieties, whereas the model assigns such reallocation to the creation of new industries. For this reason, I interpret the panel as capturing the qualitative life-cycle pattern rather than its empirical slope. Consistent with the evidence in Gort and Klepper (1982), Jovanovic and MacDonald (1994), and Filson (2001), model output rises over the life cycle.

The top-right panel shows the expected change in industry prices, adjusted for wage inflation, relative to the initial level. I deflate the industry price index from Section 3 by the wage to remove aggregate inflation. Consistent with the evidence in Gort and Klepper (1982), Agarwal (1998), and Filson (2001), the resulting series falls over the life-cycle as innovation reduces unit costs and competition intensifies.

The bottom panels plot the expected number of firms and the expected innovation rate over time. Firm counts increase while innovation rates decline, consistent with the stylized patterns documented by Abernathy and Utterback (1978), Filson (2001), and Jovanovic and Wang (2025). This trajectory captures the early expansion with frequent entry, followed by consolidation and lower innovation in more mature industries. The model does not feature an explicit shakeout phase with declining firm counts as in Klepper (1996), but such a pattern could be generated by introducing a fixed operating cost. As technology gaps widen and follower profits fall, some firms would optimally exit.

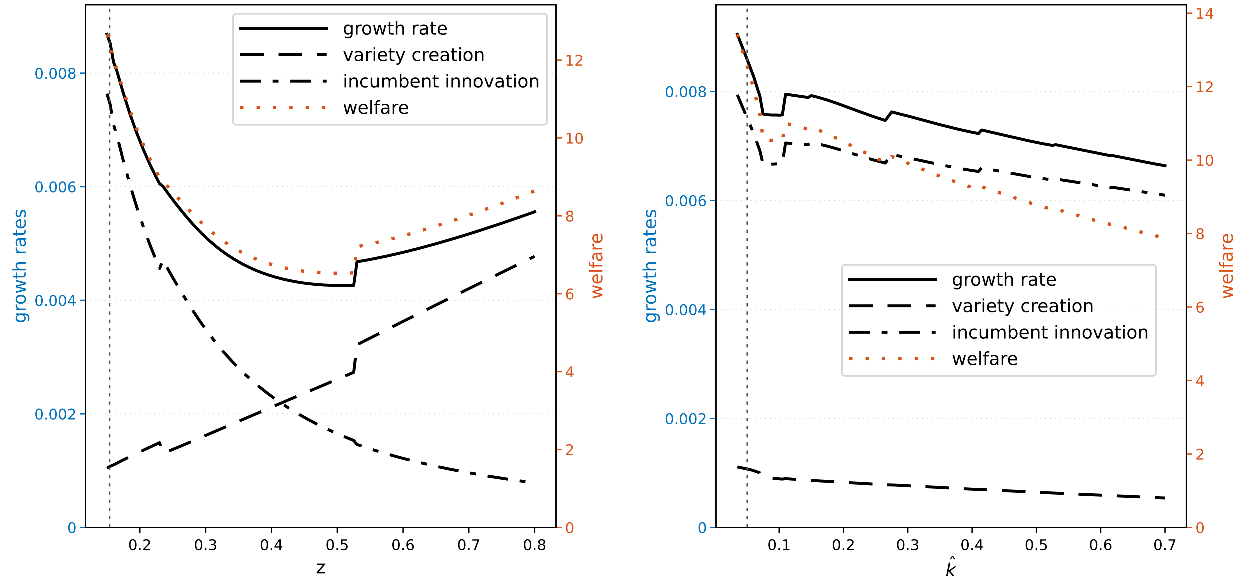
These age profiles are not directly targeted in the calibration in Table 2, so their qualitative consistency with the empirical patterns provides independent support for the model's defining feature. In addition, I verify numerically that leader innovation strictly dominates the combined innovation of the followers in every non-neck-and-neck state, a condition stronger than Assumption 5.1. It follows mathematically that the cohort expected gap function $f(a)$ is strictly increasing, so the monotone life-cycle properties in Proposition 5.3(iii) hold in the quantitative model, not only as theoretical possibilities.

6.3 Comparative Statics

To understand how the cross-sectional distribution of industries transmits parameter changes into aggregate growth, I conduct comparative statics with respect to parameters governing industry creation and competition. I first vary the potential entrant arrival rate z and the market initiation cost parameter $\hat{k}$, which control the intensity and cost of entry. I then vary the elasticity of substitution σ, which governs the strength of within-industry competition.

The left panel of Figure 3 shows how aggregate growth and its components respond to

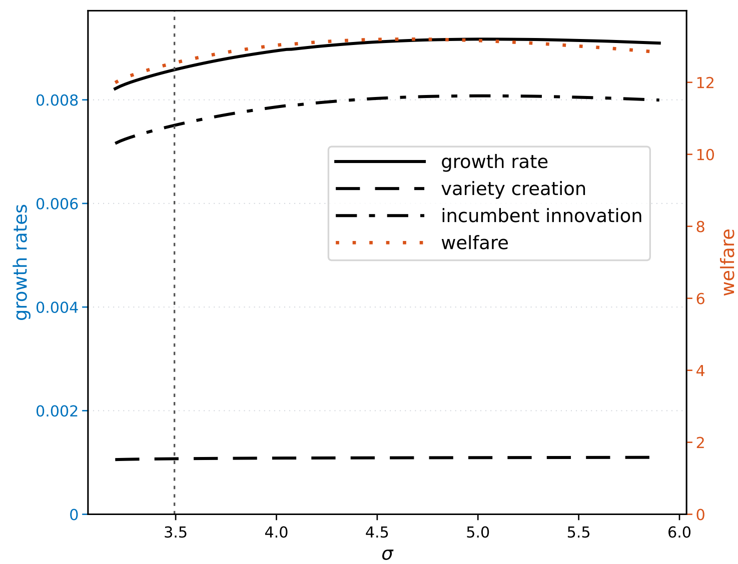
Figure 3: Comparative Statics: Changing z and $\hat{k}$



changes in z. The net result is a U-shaped relationship between z and aggregate growth. As z increases, the variety creation rate γ rises, but the average innovation rates of monopolists and oligopoly followers decline. More potential entrants generate more new industries, but they also reduce incumbent profits through both between- and within-industry business-stealing effects, lowering the expected value of innovation and crowding out R&D investment by incumbent firms. When z is low, increasing it reduces growth because the crowding-out effect dominates. Further increases beyond a threshold raise growth as the extensive-margin benefit from greater variety creation outweighs the loss from reduced incumbent innovation. The discrete jumps visible in both panels of Figure 3 reflect threshold switches in state-contingent entry decisions, including corner solutions induced by the bounded support of market-initiation costs and discrete changes in the active follower cutoff, and are discussed in Appendix C.4.

The right panel varies the mean market initiation cost $\hat{k}$. Lowering $\hat{k}$ makes new-industry creation more attractive and raises the variety creation rate γ. This extensive-margin response increases aggregate growth directly by creating more new industries. It also changes the allocation of potential entrants across margins. When creating a new industry becomes cheaper, more entrants choose the extensive margin rather than joining incumbent industries. This reduces within-industry crowding and business stealing, which raises incumbents' incentives to innovate. The opposing force is that a higher γ shortens incumbents' effective horizon by increasing creative destruction. In the calibrated economy, however, the reduction in within-industry entry pressure dominates, so incumbent innovation rises as $\hat{k}$ falls.

Figure 4: Comparative Statics: Changing σ



Growth therefore rises mainly because lower market initiation costs expand the flow of new industries and, in this calibration, also strengthen incumbent R&D incentives by redirecting entry away from existing industries.

These experiments also have a direct policy interpretation. The U-shaped relationship between z and growth cautions against the view that more entrepreneurs are always the right response to low growth. In the calibrated economy, z already lies on the downward-sloping segment of this relationship. An indiscriminate increase in the mass of potential entrants would therefore reduce growth, as additional entry mainly intensifies business stealing in existing industries and crowds out high-value incumbent R&D while yielding only limited gains in variety creation. What matters for long-run growth is not just the quantity of startups but their composition. Growth is highest when entry takes the form of new-industry creation rather than incremental entry into incumbent markets. Reducing the effective initiation cost $\hat{k}$ is a more targeted way to tilt entry toward new industries and raise growth, but it is also harder to implement in practice. The parameter $\hat{k}$ reflects deep features of the business environment, including regulation, fixed setup costs, and institutional barriers, that are difficult to change quickly.

Figure 4 reports comparative statics for the elasticity of substitution σ. The response of incumbent innovation is non-monotone. At low levels of σ, stronger product-market competition raises incumbent R&D by increasing the value of escaping close competition. This effect is strongest in industries with small technology gaps, where leaders and followers remain close competitors and the prize from moving ahead is large. As σ increases further, however, the Schumpeterian force becomes stronger. Competition compresses current profits and reduces the value of future rents, especially in industries with larger gaps or more crowded follower tiers. In these states, follower catch-up becomes less attractive and leader innovation also weakens. As a result, incumbent innovation first rises and then falls.

The aggregate growth rate follows a similar hump-shaped pattern. At moderate levels of σ, stronger competition raises growth through both incumbent innovation and the entry margin. By making entry into mature incumbent industries less attractive, it redirects some potential entrants toward the creation of new industries, increasing the contribution of variety creation. At higher levels of σ, these pro-growth forces weaken. The decline in appropriable rents reduces incumbent R&D incentives, and the induced change in the stationary distribution places more weight on states with lower innovation returns. Growth therefore eventually declines even though competition remains stronger.

This pattern echoes the inverted-U relationship between competition and innovation in Aghion et al. (2005), but the mechanism is richer in the present environment. In their framework, the effect of competition depends on firms' distance to the technological frontier. Here, competition also changes the allocation of entry between incumbent and new industries and reshapes the stationary distribution of industries over life-cycle states. The inverted-U pattern therefore reflects not only state-level innovation incentives, but also the endogenous composition of industries and the extensive margin of variety creation.

6.4 Constrained Efficiency of Calibrated Economy

I now use the constrained planner framework of Section 5.3 to assess the efficiency of variety creation in the calibrated economy. The goal is to quantify the wedge between the decentralized entry cutoffs $\{\hat{\kappa}_S^{CE}\}_S$ and the planner's cutoffs $\{\hat{\kappa}_S^{SP}\}_S$ that maximize welfare subject to the equilibrium laws of motion.

The planner's first-order condition for the cutoff in state S, equation (14), combines three components. The first is a resource cost effect through the shadow value of aggregate consumption $1/(1-\mathcal{R}-\mathcal{K})$. The second is a knowledge effect through the term $(\ln\lambda)/\rho$. The third is a composition term Φ_S that captures how reallocating entrants across states affects the cross-sectional distribution of industries and incumbent innovation incentives. In this subsection, I evaluate these objects at the competitive equilibrium.

The resource cost effect is modest but non-trivial. By construction, the planner attaches a higher utility cost to market initiation than private entrants. The shadow value of aggregate consumption is scaled by $1/(1-\mathcal{R}-\mathcal{K}) > 1$, so any given resource cost magnified relative to the private valuation. In the calibrated economy, the planner's effective valuation of the market initiation cost is 2.74% higher than the private valuation for a given cutoff $\hat{\kappa}$.

The knowledge component pushes in the opposite direction and dominates on the quantitatively relevant part of the state space. Although the private value v_1^M exceeds the simple bound $\pi_1^M/(\rho+\gamma)$ because it includes the option value of future innovation, the net private incentive $v_1^M - v_{S+}^f$ remains strictly below the social benefit $(\ln\lambda)/\rho$ for all states with economically meaningful stationary mass, defined here as $\mu_S > 0.1\%$. Ignoring higher-order composition effects, the planner's cutoff would satisfy

$$\hat{\kappa}^{SP} = \frac{1-\mathcal{R}-\mathcal{K}}{\rho}\ln\lambda.$$

Under the current calibration, this expression evaluates to 0.4830. In the model's cost normalization, this corresponds to $G(\hat{\kappa}^{SP}) = 0.9831$, so the planner would be willing to admit roughly 98% of potential market-initiation draws. This cutoff lies far above the competitive cutoff for much of the relevant state space. Conditional on holding innovation rates and industry shares fixed, the constrained-efficient allocation therefore calls for substantially more variety creation than the competitive equilibrium.

To quantify the full equilibrium wedge, I compute the derivative of welfare with respect to each state-contingent creation threshold at the competitive equilibrium. Evaluating the planner's first-order condition at the decentralized cutoffs and numerically approximating the derivatives, I find that the derivative is positive for every state with stationary mass above 2.56%. The states in which the derivative is non-positive are individually small and together account for 18.35% of stationary mass. Thus, the welfare gain from expanding variety creation is not positive at every point of the finite state space, but it is positive for all quantitatively important states and for most of the stationary distribution.

I also separate the direct externalities from the equilibrium adjustment effect. When both innovation rates and industry shares are held fixed at their competitive values, the non-positive states account for only 1.20% of stationary mass, and the largest such state has mass 0.06%. This indicates that the direct knowledge and creative-destruction externalities strongly favor additional variety creation. Allowing innovation rates or industry shares to adjust introduces some low-mass exceptions, reflecting the fact that changing creation thresholds also changes incumbent innovation incentives and the composition of industries. Nevertheless, in the full equilibrium calculation, the positive wedge remains present on all states with substantial mass.

The calibrated economy therefore exhibits too little variety creation relative to the constrained planner's benchmark on the states that matter quantitatively. Potential entrants are too willing to enter existing oligopolies to chase short-term business-stealing rents, and too reluctant to pay the fixed costs to open new markets. This distortion operates not only on the number of new industries but also on the competitive environment within existing ones, since heavier entry into mature oligopolies intensifies business stealing and weakens incumbents' innovation incentives. The policy experiments in Section 6.5 explore the quantitative importance of this wedge by comparing two instruments that operate on different margins of the innovation process, subsidies aimed at variety creation and subsidies aimed at incumbent R&D.

6.5 Policy Experiments: Variety Creation versus Incumbent R&D

The constrained efficiency analysis in Section 6.4 established that the calibrated economy produces too little variety creation. I use that wedge here to compare two policy instruments that operate on different margins of the innovation process. Both policies are financed through lump-sum taxes on the representative household:

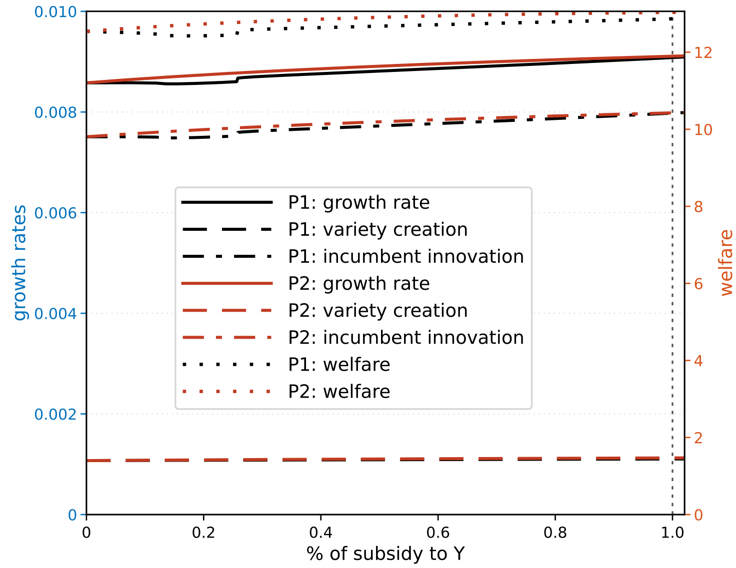
1. *Policy 1 (P1): Rewarding successful R&D investment.* A firm receives a subsidy equal to $b_r \Delta V$ upon a successful innovation, where $b_r \in [0, 1)$ is the subsidy rate and ΔV is the value increase resulting from the innovation.

2. *Policy 2 (P2): Rewarding variety creation.* A potential entrant receives a subsidy equal to $b_i v_1^M$ if it chooses to start a new industry, where $b_i \in [0, 1)$ is the subsidy rate and v_1^M is the value of a new monopolist.

Policy 1 directly targets incumbent R&D, aiming to boost technological improvement within existing industries. Policy 2 instead targets the extensive margin by increasing the returns to successful variety creation.

Figure 5 shows comparative statics for both policies across a range of subsidy-to-output ratios. In the current calibration, the growth response comes mainly through incumbent innovation rather than through variety creation. The variety-creation contribution changes only modestly under either policy, while the incumbent-innovation contribution rises steadily with the subsidy rate. Although Policy 2 is targeted at the market-creation margin, its effect on growth therefore operates primarily through the equilibrium response of incumbent R&D. By making new-industry creation more attractive than joining an existing industry, Policy 2 weakens within-industry entry pressure and raises incumbents' innovation incentives. Policy 1 raises incumbent innovation more directly, by subsidizing successful incumbent R&D. In both cases, the growth increase comes mostly from the intensive margin of incumbent innovation.

Figure 5: Policy Comparison: Rewarding Incumbent R&D and Variety Creation



To gauge the relative strength of the two instruments rather than to identify an optimal policy, I compare them at a common fiscal cost, choosing subsidy rates such that government expenditures on each policy amount to 1% of output in the stationary equilibrium. Under Policy 1, this corresponds to rewarding $b_r \approx 0.228$ of the value increase from a successful incumbent innovation. At this level, the long-run growth rate rises by about 5.02 basis points and welfare increases by 0.82%. Under Policy 2, the same fiscal-cost constraint implies a smaller subsidy rate, $b_i \approx 0.097$, applied to the value of a new monopolist. Even though variety creation itself responds only modestly, Policy 2 has larger effects in the calibrated economy. The growth rate increases by about 5.29 basis points and welfare rises by 1.25%.

The life-cycle structure of the model helps explain this response. In mature industries, additional follower entry intensifies business stealing and lowers the value of incumbent positions. A subsidy to market creation changes the outside option of potential entrants and redirects entry pressure away from existing industries. This raises the value of incumbent innovation, especially in states where firms remain close enough for R&D incentives to be responsive. The policy therefore affects growth not only by changing the number of new industries created, but also by changing the competitive environment faced by incumbent firms. In the current calibration, the latter channel is quantitatively more important.

More broadly, these results illustrate the distinctive policy margin created by endogenous industry dynamics. In fixed-industry models, innovation policy operates only through incumbent R&D. Here, policy can also change the allocation of entry between existing and new industries, and this allocation feeds back into incumbent innovation incentives. The quantitative results show that the entry margin is undervalued in equilibrium, but its main effect in this calibration is transmitted through incumbent R&D rather than through a large increase in the rate of variety creation.

7 Conclusion

This paper shows how endogenous industry life-cycles shape aggregate growth and create policy margins that models with a fixed set of industries cannot produce. Industries are born as monopolies, attract followers who compete in patent races, and mature as technology gaps widen and innovation incentives weaken. Individual industries follow non-stationary paths, but their cross-sectional distribution over life-cycle states is stationary in equilibrium, so steady aggregate growth coexists with continual turnover of the industries that drive it. This stationary distribution maps directly into the growth rate through an explicit decomposition into variety creation, frontier innovation, and catch-up innovation. Because the distribution is itself an equilibrium object, it responds to policy and to the conditions that govern entry, and that response drives the paper's other results.

Because private entrants cannot fully appropriate the knowledge that a new industry adds, the decentralized economy creates too few industries relative to the constrained optimum. This inefficiency points to a policy lever that fixed-industry frameworks do not have, the allocation of entry between existing and new industries. Equal-cost experiments calibrated to U.S. data show that the lever is quantitatively important. At a common fiscal cost, a subsidy to variety creation raises growth and welfare by more than a subsidy to incumbent R&D in the calibrated economy. The gain comes mostly through incumbent innovation. Making new industries a more attractive destination for entrants relieves entry pressure on mature industries and raises the value of incumbent positions, rather than sharply increasing the rate of variety creation.

The calibrated model also reveals how the endogenous cross-section reshapes the response of growth to entry and to competition. The relationship between startup entry and growth is U-shaped, because additional entrants first crowd out incumbent innovation and then, beyond a threshold, create enough new industries to dominate. Stronger product-market competition has a similarly non-monotone effect, raising growth at moderate levels and lowering it once it erodes the rents that fund innovation. The model also produces non-stationary industry life-cycles, with front-loaded innovation, declining entry, falling prices, and rising output, which match the stylized facts without being targeted in the calibration.

The framework also points to natural extensions. Higher market-initiation costs reduce variety creation, slow growth, and shift the industry distribution toward older, less innovative states, so a non-stationary version in which these costs evolve over time could help quantify how changing entry conditions contributed to the decline in U.S. business dynamism. The link between policy and the endogenous distribution of industries also motivates characterizing the optimal mix of incumbent R&D and industry-creation subsidies, which would clarify how the optimal intervention depends on the economy's composition of young and mature industries. More broadly, understanding long-run growth requires taking seriously the extensive margin of industry creation, not only the intensive margin of R&D within existing markets.

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A Proofs for Theoretical Results

A.1 Proof of Lemma 4.1

Proof. Since the final-good aggregator is Cobb-Douglas across industries, each industry receives expenditure $\bar{Y}$.

First consider a monopoly industry born at date τ. The monopolist has productivity $\lambda^{s_\tau^K+\hat{s}}$ and the competitive fringe has $\lambda^{s_\tau^K}$. Since the monopolist has lower marginal cost, it charges the fringe marginal cost,

$$p_{\hat{s}}^M = \frac{w}{\lambda^{s_\tau^K}},$$

and captures industry demand. Thus

$$\Pi_{\hat{s}}^M = \left(1-\lambda^{-\hat{s}}\right)\bar{Y}, \qquad \pi_{\hat{s}}^M = 1-\lambda^{-\hat{s}} \nearrow 1, \qquad \pi_{\hat{s}+1}^M - \pi_{\hat{s}}^M = (1-\lambda^{-1})\lambda^{-\hat{s}} \searrow 0.$$

Now consider an oligopoly with gap m and n followers. Followers have productivity λ^{s_f} and the leader has productivity λ^{s_f+m}. Given prices, firm j faces demand

$$y_j = \frac{p_j^{-\sigma}}{\sum_k p_k^{1-\sigma}}\bar{Y}.$$

Let $\phi_{mn} \equiv p_{mn}/p_{-mn}$. The pricing FOC gives

$$\frac{p_{mn}}{w/\lambda^{s_f+m}} = \frac{\sigma n + \phi_{mn}^{1-\sigma}}{(\sigma-1)n}, \qquad \frac{p_{-mn}}{w/\lambda^{s_f}} = \frac{\sigma(\phi_{mn}^{1-\sigma}+n-1)+1}{(\sigma-1)(\phi_{mn}^{1-\sigma}+n-1)}.$$

Taking the ratio gives

$$\phi_{mn} n \lambda^m \left(\sigma + \frac{1}{\phi_{mn}^{1-\sigma}+n-1}\right) = \sigma n + \phi_{mn}^{1-\sigma}. \tag{A.1}$$

The left-hand side is increasing in ϕ_{mn} and the right-hand side is decreasing, so ϕ_{mn} is uniquely determined by (m,n). Evaluating (A.1) at $\phi = 1$ implies $\phi_{mn} \leq 1$, with strict inequality if $m > 0$.

Define

$$a_{mn} \equiv \phi_{mn}^{1-\sigma}.$$

Then (A.1) becomes

$$\lambda^m a_{mn}^{-1/(\sigma-1)} \left(\sigma + \frac{1}{a_{mn}+n-1}\right) = \sigma + \frac{a_{mn}}{n}. \tag{A.2}$$

The normalized profits are

$$\pi_{mn} = \frac{a_{mn}}{\sigma n + a_{mn}}, \qquad \pi_{-mn} = \frac{1}{\sigma(a_{mn} + n - 1) + 1}.$$

Fix n. In (A.2), for any fixed a, the left-hand side increases with m, while the right-hand side is unchanged. Since the left-hand side decreases in a and the right-hand side increases in a, a_{mn} increases with m. Therefore π_{mn} increases and π_{-mn} decreases with m. Moreover, $a_{mn} \to \infty$ as $m \to \infty$, so

$$\pi_{mn} \nearrow 1, \qquad \pi_{-mn} \searrow 0.$$

Next fix m and let n vary. Let $r_n \equiv a_{mn}/n$. Then

$$\lambda^m (n r_n)^{-1/(\sigma-1)} \left(\sigma + \frac{1}{n(1 + r_n) - 1}\right) = \sigma + r_n. \tag{A.3}$$

For fixed r_n, the left-hand side decreases with n; for fixed n, it decreases with r_n, while the right-hand side increases with r_n. Hence r_n decreases with n. Since

$$\pi_{mn} = \frac{r_n}{\sigma + r_n},$$

leader profit decreases with n. Moreover, $r_n \to 0$, so $\pi_{mn} \searrow 0$.

It remains to show that follower profits also decrease with n. Rewrite (A.1) as

$$\lambda^m \phi = \frac{\sigma + \phi^{1-\sigma}/n}{\sigma + 1/(\phi^{1-\sigma} + n - 1)}. \tag{A.4}$$

For fixed $\phi \le 1$, let $A \equiv \phi^{1-\sigma} \ge 1$ and define

$$H(n; A) \equiv \frac{\sigma + A/n}{\sigma + 1/(A + n - 1)}.$$

Treating n continuously, $H(n; A)$ is weakly decreasing in n. Indeed, $\partial H(n; A)/\partial n \le 0$ is equivalent to

$$A(A + n - 1)\big[\sigma(A + n - 1) + 1\big] \ge n(\sigma n + A),$$

whose left-hand side minus right-hand side equals

$$(A - 1)\left(A^2\sigma + 2An\sigma - A\sigma + A + n^2\sigma\right) \ge 0.$$

Thus, holding ϕ fixed, the right-hand side of (A.4) decreases with n. Since the left-hand side of (A.4) is increasing in ϕ, while the right-hand side is decreasing in ϕ, the equilibrium solution ϕ_{mn} weakly decreases with n. Therefore $a_{mn} = \phi_{mn}^{1-\sigma}$ weakly increases with n, and hence $a_{mn} + n - 1$ increases with n. It follows that

$$\pi_{-mn} = \frac{1}{\sigma(a_{mn} + n - 1) + 1}$$

decreases with n. Since $a_{mn} + n - 1 \to \infty$ as $n \to \infty$,

$$\pi_{-mn} \searrow 0.$$

Finally, in a monopoly,

$$\frac{P^M(\tau)}{w} = \lambda^{-s_\tau^K},$$

so $P^M(\tau)/w \searrow 0$. In an oligopoly, with $b_{mn} \equiv a_{mn} + n - 1$,

$$\frac{P_{mn}(s_f)}{w} = \lambda^{-s_f} \frac{\sigma b_{mn} + 1}{(\sigma - 1)b_{mn}} (b_{mn} + 1)^{1/(1-\sigma)}.$$

The term multiplying λ^{-s_f} is decreasing in b_{mn}. Since b_{mn} increases with both m and n, the wage-adjusted industry price decreases in m, n, and s_f. Moreover, $b_{mn} \to \infty$ as $m \to \infty$ or $n \to \infty$, and $\lambda^{-s_f} \to 0$ as $s_f \to \infty$. Hence

$$\frac{P_{mn}(s_f)}{w} \searrow 0$$

as $m \to \infty$, $n \to \infty$, or $s_f \to \infty$. □

A.2 Proof of Lemma 4.2

Proof. Let

$$\Phi(d) \equiv \max_{x \geq 0} \{-R(x) + xd\}.$$

Under the maintained assumptions on R, Φ is increasing, $\Phi(d) = 0$ for $d \leq 0$, and Φ is strictly increasing for $d > 0$. The optimal innovation rate is increasing in d and is strictly positive if and only if $d > 0$.

Part (i). By Lemma 4.1, normalized flow profits are strictly between zero and one. Since R&D costs are nonnegative,

$$v < \int_0^\infty e^{-(\rho+\gamma)t} dt = \frac{1}{\rho + \gamma}.$$

The lower bound follows because a firm can choose zero R&D and receive strictly positive flow profits while the industry is active. Hence

$$0 < v < \frac{1}{\rho+\gamma}.$$

Part (ii). The monopoly HJB is

$$(\rho+\gamma+z\theta_M)v_{\hat{s}}^M = \pi_{\hat{s}}^M + \Phi(v_{\hat{s}+1}^M - v_{\hat{s}}^M) + z\theta_M v_{01}.$$

Suppose $v_{\hat{s}+1}^M \leq v_{\hat{s}}^M$. Then $\Phi(v_{\hat{s}+1}^M - v_{\hat{s}}^M) = 0$, so

$$(\rho+\gamma+z\theta_M)v_{\hat{s}}^M = \pi_{\hat{s}}^M + z\theta_M v_{01}.$$

But

$$(\rho+\gamma+z\theta_M)v_{\hat{s}+1}^M = \pi_{\hat{s}+1}^M + \Phi(v_{\hat{s}+2}^M - v_{\hat{s}+1}^M) + z\theta_M v_{01} > \pi_{\hat{s}}^M + z\theta_M v_{01},$$

contradicting $v_{\hat{s}+1}^M \leq v_{\hat{s}}^M$. Thus $v_{\hat{s}}^M$ is strictly increasing.

Since $\{v_{\hat{s}}^M\}$ is increasing and bounded, it converges to some L, with $v_{\hat{s}+1}^M - v_{\hat{s}}^M \to 0$. Taking limits in the HJB and using $\pi_{\hat{s}}^M \to 1$ gives

$$L = \frac{1+z\theta_M v_{01}}{\rho+\gamma+z\theta_M}.$$

Also, $v_{\hat{s}+1}^M - v_{\hat{s}}^M > 0$ implies $x_{\hat{s}}^M > 0$, while $v_{\hat{s}+1}^M - v_{\hat{s}}^M \to 0$ implies $x_{\hat{s}}^M \to 0$.

It remains to show that $x_{\hat{s}}^M$ decreases. Define

$$\Delta v_{\hat{s}}^M \equiv v_{\hat{s}+1}^M - v_{\hat{s}}^M, \qquad \Delta \pi_{\hat{s}}^M \equiv \pi_{\hat{s}+1}^M - \pi_{\hat{s}}^M.$$

Subtracting the HJB at $\hat{s}$ from the HJB at $\hat{s}+1$ gives

$$(\rho+\gamma+z\theta_M)\Delta v_{\hat{s}}^M = \Delta\pi_{\hat{s}}^M + \Phi(\Delta v_{\hat{s}+1}^M) - \Phi(\Delta v_{\hat{s}}^M).$$

By Lemma 4.1, $\Delta\pi_{\hat{s}}^M$ is decreasing. If

$$\Delta v_{\hat{s}}^M > \Delta v_{\hat{s}+1}^M \quad \text{and} \quad \Delta v_{\hat{s}-1}^M \leq \Delta v_{\hat{s}}^M,$$

then

$$\begin{aligned}(\rho+\gamma+z\theta_M)\Delta v_{\hat{s}}^M &= \Delta\pi_{\hat{s}}^M + \Phi(\Delta v_{\hat{s}+1}^M) - \Phi(\Delta v_{\hat{s}}^M) \\ &< \Delta\pi_{\hat{s}-1}^M + \Phi(\Delta v_{\hat{s}}^M) - \Phi(\Delta v_{\hat{s}-1}^M) \\ &= (\rho+\gamma+z\theta_M)\Delta v_{\hat{s}-1}^M,\end{aligned}$$

a contradiction. Hence any strict decline in $\Delta v_{\hat{s}}^M$ propagates backward. Since $\Delta v_{\hat{s}}^M > 0$ and $\Delta v_{\hat{s}}^M \to 0$, strict declines occur arbitrarily far out. Therefore

$$\Delta v_{\hat{s}}^M > \Delta v_{\hat{s}+1}^M \qquad \text{for all } \hat{s}.$$

Since the innovation rate is increasing in $\Delta v_{\hat{s}}^M$, $x_{\hat{s}}^M$ strictly decreases in $\hat{s}$.

Part (iii). By part (i), all values are bounded by $\frac{1}{\rho+\gamma}$, so all individual innovation rates are bounded above by some $\bar{x} < \infty$.

Fix m and let $n \to \infty$. For $K \geq 1$, define

$$\tau_K \equiv \inf\{t \geq 0 : \text{the gap exceeds } m + K\}.$$

Before τ_K, the gap is at most $m+K$ and the number of followers is at least n. By Lemma 4.1, both leader and follower flow profits are therefore bounded above by $\pi_{m+K,n}$, and for each fixed K,

$$\pi_{m+K,n} \to 0 \qquad \text{as } n \to \infty.$$

After τ_K, continuation values are bounded by $\frac{1}{\rho+\gamma}$. Dropping R&D costs,

$$\max\{v_{mn}, v_{-mn}\} \leq \frac{\eta_{nK}}{\rho+\gamma} + \frac{1}{\rho+\gamma}\,\mathbb{E}\left[e^{-r\tau_K}\right].$$

To reach a gap above $m+K$, the leader must obtain at least K successful innovations. Since the leader's innovation rate is bounded by $\bar{x}$,

$$\mathbb{E}\left[e^{-r\tau_K}\right] \leq \left(\frac{\bar{x}}{r+\bar{x}}\right)^K.$$

Therefore,

$$\limsup_{n\to\infty} \max\{v_{mn}, v_{-mn}\} \leq \frac{1}{\rho+\gamma}\left(\frac{\bar{x}}{r+\bar{x}}\right)^K.$$

Letting $K \to \infty$ gives

$$v_{mn} \to 0, \qquad v_{-mn} \to 0 \qquad \text{as } n \to \infty.$$

Since the relevant value gains also converge to zero,

$$x_{mn} \to 0, \qquad x_{-mn} \to 0 \qquad \text{as } n \to \infty.$$

Now fix n and let $m \to \infty$. We first show that $v_{-mn} \to 0$. For an integer $\ell < m$, define

$$\tau_\ell \equiv \inf\{t \geq 0 : \text{the gap is at most } \ell\}.$$

Before τ_ℓ, the gap is larger than ℓ and the number of followers is at least n. By Lemma 4.1, follower flow profits are therefore bounded above by $\pi_{-\ell n}$, and

$$\pi_{-\ell n} \to 0 \qquad \text{as } \ell \to \infty.$$

Thus, dropping R&D costs and using the bound on continuation values,

$$v_{-mn} \leq \frac{\pi_{-\ell n}}{\rho + \gamma} + \frac{1}{\rho + \gamma} \mathbb{E}\left[e^{-r\tau_\ell}\right].$$

For fixed ℓ, reaching gap ℓ from gap m requires at least $m - \ell$ follower innovations. Since individual innovation rates are bounded and entrant arrivals occur at rate at most z,

$$\mathbb{E}\left[e^{-r\tau_\ell}\right] \to 0 \qquad \text{as } m \to \infty.$$

Hence

$$\limsup_{m \to \infty} v_{-mn} \leq \frac{\pi_{-\ell n}}{\rho + \gamma}.$$

Letting $\ell \to \infty$ gives

$$v_{-mn} \to 0.$$

Since both v_{-mn} and $v_{-(m-1)n}$ converge to zero,

$$x_{-mn} \to 0.$$

Finally consider the leader. Since $v_{-m,n+1} \to 0$ and $\overline{\kappa} < v_1^M$, entrants eventually strictly prefer creating a new industry to joining a sufficiently wide-gap oligopoly. Hence $\theta_{mn} = 0$ for all sufficiently large m. By Lemma 4.1, $\pi_{mn} \to 1$. Choosing zero R&D in the leader's HJB gives

$$(\rho + \gamma)v_{mn} \geq \pi_{mn} + n x_{-mn}(v_{m-1,n} - v_{mn}) + z\theta_{mn}(v_{m,n+1} - v_{mn}).$$

The last two terms vanish because values are bounded, $x_{-mn} \to 0$, and $\theta_{mn} = 0$ eventually. Hence

$$\liminf_{m \to \infty} v_{mn} \geq \frac{1}{\rho + \gamma}.$$

Together with part (i), this implies

$$v_{mn} \to \frac{1}{\rho + \gamma}.$$

Since v_{mn} and $v_{m+1,n}$ converge to the same limit, $x_{mn} \to 0$.

Part (iv). From part (iii), $v_{0n} \to 0$ as $n \to \infty$. Since $\overline{\kappa} < v_1^M$, choose N large enough such that

$$\sup_{n' \geq N} v_{0n'} < v_1^M - \overline{\kappa}, \qquad \sup_{n' \geq N} \frac{\pi_{0n'}}{\rho + \gamma} < v_1^M - \overline{\kappa}.$$

Consider a follower in state (m, n) with $n \geq N$. Before reaching neck-and-neck, its flow payoff is bounded above by π_{0n} because Lemma 4.1 implies

$$\pi_{-m'n'} \leq \pi_{0n'} \leq \pi_{0n} \qquad \text{for all } m' \geq 0,\ n' \geq n.$$

After reaching neck-and-neck, its continuation value is bounded above by $\sup_{n' \geq n} v_{0n'}$. Hence

$$v_{-mn} \leq \max \left\{ \frac{\pi_{0n}}{\rho + \gamma}, \sup_{n' \geq n} v_{0n'} \right\} < v_1^M - \overline{\kappa}.$$

Thus, for any state with $n \geq N$ followers,

$$v_{-m,n+1} < v_1^M - \overline{\kappa}.$$

Even an entrant with the highest creation cost $\overline{\kappa}$ strictly prefers creating a new industry to joining. Therefore there exists a finite $n^* \geq 1$ such that

$$\theta_{mn} = 0 \qquad \text{for all } m \text{ and all } n \geq n^*.$$

$\square$

A.3 Proof of Lemma 4.4

Proof. In a stationary equilibrium, $\delta + g_N = \gamma$, and the average levels satisfy

$$\begin{aligned}
\mu_M \dot{\bar{s}}_t^M &= -(z\theta_M + \gamma)\mu_M \bar{s}_t^M + \gamma(s_t^K + 1) + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M, \\
\mu_M \dot{\bar{s}}_t^{cf} &= -(z\theta_M + \gamma)\mu_M \bar{s}_t^{cf} + \gamma s_t^K, \\
(1-\mu_M)\dot{\bar{s}}_t^f &= -\gamma(1-\mu_M)\bar{s}_t^f + z\theta_M \mu_M \bar{s}_t^M + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}, \\
(1-\mu_M)\dot{\bar{s}}_t^l &= -\gamma(1-\mu_M)\bar{s}_t^l + z\theta_M \mu_M \bar{s}_t^M + \sum_{n=1}^{n^*}\left[(n+1)x_{0n}\mu_{0n} + \sum_{m=1}^{\infty} x_{mn}\mu_{mn}\right].
\end{aligned}$$

Using $s_t^K = \mu_M \bar{s}_t^M + (1-\mu_M)\bar{s}_t^f$, the first and third equations imply

$$\dot{s}_t^K = \gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}.$$

Next, subtracting the second equation from the first gives

$$\mu_M \frac{d}{dt}(\bar{s}_t^M - \bar{s}_t^{cf}) = -(z\theta_M + \gamma)\mu_M(\bar{s}_t^M - \bar{s}_t^{cf}) + \gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M.$$

Since

$$\bar{s}_t^M - \bar{s}_t^{cf} = \frac{1}{\mu_M}\sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M \hat{s}$$

is constant in stationarity and

$$\mu_M = \frac{\gamma}{z\theta_M + \gamma},$$

we obtain

$$\gamma(\bar{s}_t^M - \bar{s}_t^{cf}) = \gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M.$$

Finally, subtracting the third equation from the fourth gives

$$(1-\mu_M)\frac{d}{dt}(\bar{s}_t^l - \bar{s}_t^f) = -\gamma(1-\mu_M)(\bar{s}_t^l - \bar{s}_t^f) + \sum_{n=1}^{n^*}\left[(n+1)x_{0n}\mu_{0n} + \sum_{m=1}^{\infty} x_{mn}\mu_{mn}\right] - \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}.$$

Since

$$\bar{s}_t^l - \bar{s}_t^f = \bar{m} = \frac{1}{1-\mu_M}\sum_{n=1}^{n^*}\sum_{m=0}^{\infty} \mu_{mn} m$$

is constant in stationarity,

$$\gamma(1-\mu_M)(\bar{s}^l_t-\bar{s}^f_t)=\gamma(1-\mu_M)\bar{m}=\sum_{n=1}^{n^*}\left[(n+1)x_{0n}\mu_{0n}+\sum_{m=1}^{\infty}x_{mn}\mu_{mn}\right]-\sum_{n=1}^{n^*}\sum_{m=1}^{\infty}nx_{-mn}\mu_{mn}.$$

$\square$

A.4 Proof of Proposition 5.1

Proof. A monopoly born at time τ has output

$$Y^M(\hat{s},\tau)=\frac{\bar{Y}_t}{w_t}\lambda^{s^K_\tau},$$

and an oligopoly with follower technology s^f has output

$$Y_{mn}(s^f)=\tilde{Y}^f_{mn}\lambda^{s^f}\frac{\bar{Y}_t}{w_t},$$

where $\tilde{Y}^f_{mn}$ depends only on (m,n). Since

$$\bar{Y}_t=\exp\left(\frac{1}{N_t}\int_0^{N_t}\ln Y_{it}\,di\right),$$

these output expressions imply

$$\ln w_t=\ln\lambda\left[\mu_M\bar{s}^{cf}_t+(1-\mu_M)\bar{s}^f_t\right]+\sum_{n=1}^{n^*}\sum_{m=0}^{\infty}\mu_{mn}\ln\tilde{Y}^f_{mn}.$$

The last term is constant in a stationary equilibrium, so

$$\dot{\ln w_t}=\ln\lambda\left[\mu_M\dot{\bar{s}}^{cf}_t+(1-\mu_M)\dot{\bar{s}}^f_t\right].$$

The labor-market condition implies that $\bar{Y}_t/w_t$ grows at rate $-g_N$. Hence

$$g_t=\dot{\ln Y}_t=g_N+\dot{\ln\bar{Y}_t}=\dot{\ln w}_t.$$

Therefore

$$g_t=\ln\lambda\left[\mu_M\dot{\bar{s}}^{cf}_t+(1-\mu_M)\dot{\bar{s}}^f_t\right].$$

From the level equations in Lemma 4.4,

$$\mu_M \dot{\bar{s}}_t^{cf} = -(z\theta_M + \gamma)\mu_M \bar{s}_t^{cf} + \gamma s_t^K,$$
$$(1-\mu_M)\dot{\bar{s}}_t^{f} = -\gamma(1-\mu_M)\bar{s}_t^{f} + z\theta_M \mu_M \bar{s}_t^M + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}.$$

Adding them and using

$$s_t^K = \mu_M \bar{s}_t^M + (1-\mu_M)\bar{s}_t^f, \qquad \mu_M = \frac{\gamma}{z\theta_M + \gamma},$$

gives

$$\mu_M \dot{\bar{s}}_t^{cf} + (1-\mu_M)\dot{\bar{s}}_t^{f} = \gamma(\bar{s}_t^M - \bar{s}_t^{cf}) + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}.$$

By Lemma 4.4,

$$\gamma(\bar{s}_t^M - \bar{s}_t^{cf}) = \gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M.$$

Thus

$$g = \ln\lambda \left(\gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M + \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}\right).$$

The equality $g = \ln\lambda \cdot \dot{s}^K$ follows directly from Lemma 4.4.

Finally, Lemma 4.4 implies

$$\gamma(1-\mu_M)\bar{m} = \sum_{n=1}^{n^*}\left[(n+1)x_{0n}\mu_{0n} + \sum_{m=1}^{\infty} x_{mn}\mu_{mn}\right] - \sum_{n=1}^{n^*}\sum_{m=1}^{\infty} n x_{-mn}\mu_{mn}.$$

Substituting this into the expression for g yields

$$g = \ln\lambda \left(\gamma + \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M x_{\hat{s}}^M + \sum_{n=1}^{n^*}\left[(n+1)x_{0n}\mu_{0n} + \sum_{m=1}^{\infty} x_{mn}\mu_{mn}\right] - \gamma(1-\mu_M)\bar{m}\right).$$

□

A.5 Proof of Proposition 5.2

Proof. For part (i), fix an oligopoly state (m, n). If $m = 0$, any successful innovation by one of the $n+1$ neck-and-neck firms creates a one-step gap, so

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta}\mathbb{E}[\tilde{m}_\Delta - \tilde{m} \mid \tilde{m} = 0, \tilde{n} = n] = (n+1)x_{0n}.$$

If $m \geq 1$, leader innovation widens the gap by one and follower innovation narrows it by one, so

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}[\tilde{m}_\Delta - \tilde{m} \mid \tilde{m} = m, \tilde{n} = n] = x_{mn} - n x_{-mn}.$$

Taking expectations over the stationary cross-sectional distribution conditional on being an oligopoly gives

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}[\tilde{m}_\Delta - \tilde{m}] = \frac{1}{1 - \mu_M} \left[\sum_{n=1}^{n^*} (n+1) x_{0n} \mu_{0n} + \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} x_{mn} \mu_{mn} - \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} n x_{-mn} \mu_{mn} \right].$$

By Lemma 4.4, the bracketed term equals $\gamma(1 - \mu_M)\bar{m}$. Therefore

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}[\tilde{m}_\Delta - \tilde{m}] = \gamma \bar{m} > 0.$$

For part (ii), let $U \sim \mathrm{Exp}(\gamma)$ denote the age of a randomly sampled surviving oligopoly from the stationary cross-section. Then

$$\bar{m} = \mathbb{E}[f(U)].$$

For any $\varepsilon > 0$,

$$\bar{m} = (1 - e^{-\gamma\varepsilon}) \mathbb{E}[f(U) \mid U \leq \varepsilon] + e^{-\gamma\varepsilon} \bar{m}_\varepsilon.$$

A newborn oligopoly starts with zero gap. Starting from this state, the probability of any innovation event before age a is $O(a)$, so the expected gap satisfies

$$f(a) = \mathbb{E}[\hat{m}_a \mid \text{alive at age } a] \to 0 \qquad \text{as } a \downarrow 0.$$

It follows that

$$\mathbb{E}[f(U) \mid U \leq \varepsilon] \to 0 \qquad \text{as } \varepsilon \downarrow 0.$$

Since $\bar{m} > 0$, for all sufficiently small $\varepsilon > 0$,

$$\mathbb{E}[f(U) \mid U \leq \varepsilon] < \bar{m}.$$

Rearranging the mixture identity gives

$$\bar{m}_\varepsilon - \bar{m} = \frac{1 - e^{-\gamma\varepsilon}}{e^{-\gamma\varepsilon}} \left[\bar{m} - \mathbb{E}[f(U) \mid U \leq \varepsilon] \right] > 0.$$

Thus $\bar{m}_\varepsilon > \bar{m}$.

For part (iii), exit occurs at the exogenous state-independent rate δ, so conditioning on survival to age a does not change the within-oligopoly transition rates. Therefore

$$f'(a) = \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}[\hat{m}_{a+\Delta} - \hat{m}_a \mid \text{alive at age } a].$$

Under Assumption 5.1, this expected instantaneous change is weakly positive for all sufficiently large a. Hence $f'(a) \geq 0$ for all sufficiently large a, so $f(a)$ is weakly increasing for sufficiently large ages. $\square$

A.6 Proof of Proposition 5.3

Proof. For part (i), conditional on being alive and still a monopoly at age a, the monopolist's quality changes only through its own innovation. Hence

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{E}\left[\ln q^M_{a+\Delta} - \ln q^M_a \,\middle|\, \text{still monopoly, alive at age } a\right] = \ln \lambda \, \mathbb{E}\left[x^M_{\hat{s}_a} \,\middle|\, \text{still monopoly, alive at age } a\right].$$

While the industry remains a monopoly, $\hat{s}_a$ is weakly increasing along every history. Conditional on remaining a monopoly, the innovation process has strictly positive arrival rates at every finite $\hat{s}$, so $\hat{s}_a \to \infty$ almost surely as $a \to \infty$. By Lemma 4.2, $x^M_{\hat{s}}$ is decreasing in $\hat{s}$ and $\lim_{\hat{s} \to \infty} x^M_{\hat{s}} = 0$. Therefore

$$\mathbb{E}\left[x^M_{\hat{s}_a} \,\middle|\, \text{still monopoly, alive at age } a\right] \to 0 \qquad \text{as } a \to \infty,$$

and monopoly quality growth slows to zero.

Part (ii) follows directly from Lemma 4.1. The lemma shows that the wage-adjusted industry price is decreasing in the technology gap m and in the number of followers n. Since Cobb-Douglas aggregation implies fixed industry expenditure $\bar{Y}$, industry output satisfies

$$Y_{mn}(s_f) = \frac{\bar{Y}}{P_{mn}(s_f)}.$$

Therefore, a lower industry price implies higher industry output.

For part (iii), Proposition 5.2(iii) implies that, under Assumption 5.1, the expected technology gap of sufficiently mature surviving oligopoly cohorts is weakly increasing. The number of followers is nondecreasing along every industry history. Hence mature surviving oligopoly cohorts move, in this sense, toward states with larger gaps and weakly larger follower counts.

By part (ii), such states have lower relative prices and higher quantities. By Lemma 4.2,

$$\lim_{m\to\infty} x_{mn} = \lim_{m\to\infty} x_{-mn} = 0$$

for each fixed n. Moreover, as $m \to \infty$, follower values converge to zero, so joining mature large-gap industries becomes unattractive. Thus mature oligopoly states have weaker innovation incentives and lower entry incentives. Finally, leadership turnover over any fixed interval becomes less likely in large-gap states, because turnover requires followers to close the leader's gap before leadership can change, while follower catch-up rates vanish as $m \to \infty$. Therefore sufficiently mature surviving oligopoly cohorts move toward lower-price, higher-quantity, lower-innovation, lower-entry, and lower-turnover states. □

B Stationary Equilibrium Conditions

This appendix collects the equilibrium conditions omitted from the main text. The normalized Hamilton-Jacobi-Bellman equations and the potential entrant's problem are stated in the main text. I first derive the normalization of firm values, then state the entry cutoffs, Kolmogorov forward equations, and aggregate market-clearing conditions.

B.1 Normalization of Firm Values

Let

$$v \equiv \frac{V}{\bar{Y}},$$

with corresponding state subscripts and superscripts. Since

$$\dot{V} = \dot{v}\bar{Y} + v\dot{\bar{Y}}$$

and

$$\frac{\dot{\bar{Y}}}{\bar{Y}} = g - g_N = r - \rho - g_N,$$

we have

$$rV - \dot{V} = rv\bar{Y} - \dot{v}\bar{Y} - v\dot{\bar{Y}} = (\rho + g_N)v\bar{Y} - \dot{v}\bar{Y}.$$

In a stationary equilibrium, normalized values are time-invariant, so $\dot{v} = 0$. Dividing the unnormalized HJB equations by $\bar{Y}$ therefore yields the normalized HJB equations in the main text, with effective discount rate $\rho + g_N$. Equivalently, using $g_N = \gamma - \delta$, moving the exit term to the left gives the effective discount rate $\rho + \gamma$.

B.2 Entry Cutoffs and Variety Creation

Let $\underline{\kappa}$ and $\bar{\kappa}$ denote the lower and upper bounds of the industry-creation cost distribution G. For any scalar a, define

$$[a]_{\underline{\kappa}}^{\bar{\kappa}} \equiv \min\{\bar{\kappa}, \max\{\underline{\kappa}, a\}\}.$$

The industry-creation cutoff for a potential entrant who meets a monopoly is

$$\hat{\kappa}_M = [v_1^M - v_{01}]_{\underline{\kappa}}^{\bar{\kappa}}.$$

For an oligopoly with $m = 0$ and $n < n^*$, the cutoff is

$$\hat{\kappa}_{0n} = [v_1^M - v_{0,n+1}]_{\underline{\kappa}}^{\bar{\kappa}}, \qquad n = 1, \ldots, n^* - 1.$$

For an oligopoly with $m \geq 1$ and $n < n^*$, the cutoff is

$$\hat{\kappa}_{mn} = [v_1^M - v_{-m,n+1}]_{\underline{\kappa}}^{\bar{\kappa}}, \qquad m \geq 1, \quad n = 1, \ldots, n^* - 1.$$

At $n = n^*$, joining the incumbent industry is not profitable. I therefore set

$$\hat{\kappa}_{m,n^*} = \bar{\kappa}, \qquad \theta_{m,n^*} = 0, \qquad m \geq 0.$$

For any state S, the probability that a potential entrant creates a new industry is $G(\hat{\kappa}_S)$, while the probability that it joins the incumbent industry is

$$\theta_S = 1 - G(\hat{\kappa}_S).$$

Let

$$\mu^M \equiv \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^M.$$

The gross rate of new-industry creation is then

$$\gamma = z \left[G(\hat{\kappa}_M)\mu^M + \sum_{n=1}^{n^*} \sum_{m=0}^{\infty} G(\hat{\kappa}_{mn})\mu_{mn} \right].$$

B.3 Kolmogorov Forward Equations

Let $\mu_{\hat{s}}^M$ denote the share of monopoly industries with technology gap $\hat{s}$, and let μ_{mn} denote the share of oligopoly industries with technology gap m and n followers. Industry shares satisfy

$$\sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^{M} + \sum_{n=1}^{n^*} \sum_{m=0}^{\infty} \mu_{mn} = 1. \tag{15}$$

The mass of industries grows at rate

$$g_N = \gamma - \delta. \tag{16}$$

The monopoly-state shares satisfy

$$(\delta + g_N)\mu_1^M + \dot{\mu}_1^M = \gamma - x_1^M \mu_1^M - z\theta_M \mu_1^M, \tag{17}$$

$$(\delta + g_N)\mu_{\hat{s}}^M + \dot{\mu}_{\hat{s}}^M = x_{\hat{s}-1}^M \mu_{\hat{s}-1}^M - x_{\hat{s}}^M \mu_{\hat{s}}^M - z\theta_M \mu_{\hat{s}}^M, \qquad \hat{s} \geq 2. \tag{18}$$

The neck-and-neck oligopoly shares satisfy

$$(\delta + g_N)\mu_{01} + \dot{\mu}_{01} = x_{-1,1}\mu_{11} - 2x_{01}\mu_{01} + z\theta_M \mu^M - z\theta_{01}\mu_{01}, \tag{19}$$

$$(\delta + g_N)\mu_{0n} + \dot{\mu}_{0n} = n x_{-1,n}\mu_{1n} - (n+1)x_{0n}\mu_{0n} + z\theta_{0,n-1}\mu_{0,n-1} - z\theta_{0n}\mu_{0n}, \qquad n = 2, \ldots, n^*. \tag{20}$$

For positive-gap oligopolies with $m = 1$, the shares satisfy

$$(\delta + g_N)\mu_{11} + \dot{\mu}_{11} = 2x_{01}\mu_{01} + x_{-2,1}\mu_{21} - (x_{11} + x_{-1,1})\mu_{11} - z\theta_{11}\mu_{11}, \tag{21}$$

$$\begin{aligned}(\delta + g_N)\mu_{1n} + \dot{\mu}_{1n} = {} & (n+1)x_{0n}\mu_{0n} + n x_{-2,n}\mu_{2n} - (x_{1n} + n x_{-1,n})\mu_{1n} \\ & + z\theta_{1,n-1}\mu_{1,n-1} - z\theta_{1n}\mu_{1n}, \qquad n = 2, \ldots, n^*.\end{aligned} \tag{22}$$

For positive-gap oligopolies with $m \geq 2$, the shares satisfy

$$(\delta + g_N)\mu_{m1} + \dot{\mu}_{m1} = x_{m-1,1}\mu_{m-1,1} + x_{-(m+1),1}\mu_{m+1,1} - (x_{m1} + x_{-m1})\mu_{m1} - z\theta_{m1}\mu_{m1}, \qquad m \geq 2, \tag{23}$$

$$\begin{aligned}(\delta + g_N)\mu_{mn} + \dot{\mu}_{mn} = {} & x_{m-1,n}\mu_{m-1,n} + n x_{-(m+1),n}\mu_{m+1,n} - (x_{mn} + n x_{-mn})\mu_{mn} \\ & + z\theta_{m,n-1}\mu_{m,n-1} - z\theta_{mn}\mu_{mn}, \\ & m \geq 2, \quad n = 2, \ldots, n^*.\end{aligned} \tag{24}$$

In a stationary equilibrium,

$$\dot{\mu}_{\hat{s}}^M = 0, \qquad \dot{\mu}_{mn} = 0$$

for all states.

B.4 Market-Clearing Conditions

The aggregate resource constraint is

$$Y = C + Y(\mathcal{R} + \mathcal{K}), \tag{25}$$

where $\mathcal{R}$ is the output share spent on incumbent R&D and $\mathcal{K}$ is the output share spent on new-industry creation.

The incumbent R&D share is

$$\mathcal{R} = \sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^{M} R(x_{\hat{s}}^{M}) + \sum_{n=1}^{n^*} \mu_{0n}(n+1)R(x_{0n}) + \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} \mu_{mn} \left[R(x_{mn}) + nR(x_{-mn}) \right]. \tag{26}$$

The new-industry creation cost share is

$$\mathcal{K} = z \left[\mu^M \int_{\underline{\kappa}}^{\hat{\kappa}_M} \kappa \, dG(\kappa) + \sum_{n=1}^{n^*} \sum_{m=0}^{\infty} \mu_{mn} \int_{\underline{\kappa}}^{\hat{\kappa}_{mn}} \kappa \, dG(\kappa) \right]. \tag{27}$$

Let $\ell_{\hat{s}}^{M}$, ℓ_{0n}, and ℓ_{mn} denote the normalized labor demand per unit of industry revenue implied by the static pricing equilibrium in the corresponding states. Labor-market clearing is

$$1 = \frac{Y}{w} \left[\sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^{M} \ell_{\hat{s}}^{M} + \sum_{n=1}^{n^*} \mu_{0n} \ell_{0n} + \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} \mu_{mn} \ell_{mn} \right]. \tag{28}$$

Aggregate household wealth equals the total market value of incumbent firms:

$$A = Y \left[\sum_{\hat{s}=1}^{\infty} \mu_{\hat{s}}^{M} v_{\hat{s}}^{M} + \sum_{n=1}^{n^*} \mu_{0n}(n+1) v_{0n} + \sum_{n=1}^{n^*} \sum_{m=1}^{\infty} \mu_{mn} \left(v_{mn} + n v_{-mn} \right) \right]. \tag{29}$$

B.5 Equilibrium Conditions

A stationary equilibrium consists of normalized firm values

$$\{v_{\hat{s}}^{M}\}_{\hat{s} \geq 1}, \qquad \{v_{0n}\}_{n=1}^{n^*}, \qquad \{v_{mn}, v_{-mn}\}_{m \geq 1,\ n=1,\ldots,n^*},$$

innovation policies

$$\{x_{\hat{s}}^{M}\}_{\hat{s} \geq 1}, \qquad \{x_{0n}\}_{n=1}^{n^*}, \qquad \{x_{mn}, x_{-mn}\}_{m \geq 1,\ n=1,\ldots,n^*},$$

entry decisions

$$\theta_M, \qquad \{\theta_{mn}\}_{m \geq 0,\ n=1,\ldots,n^*},$$

an invariant industry distribution

$$\{\mu_{\hat{s}}^M\}_{\hat{s}\geq 1}, \qquad \{\mu_{mn}\}_{m\geq 0,\ n=1,\ldots,n^*},$$

and a balanced-growth path for aggregate variables such that:

1. Firm values and innovation policies solve the normalized HJB equations stated in the main text.

2. Potential entrants optimally choose whether to join an incumbent industry or create a new industry, generating the cutoffs and joining probabilities in Section B.2.

3. Industry shares satisfy the Kolmogorov forward equations (17)–(24) with all time derivatives equal to zero.

4. The mass of industries evolves according to $g_N = \gamma - \delta$.

5. The aggregate resource, labor-market, and asset-market clearing conditions hold.

6. The household Euler equation holds, so that along the balanced growth path

$$g = r - \rho.$$

C Quantitative Appendix

C.1 Solving the Stationary Equilibrium

I compute the stationary equilibrium on a finite state space. Monopoly states are $M_{\hat{s}}$, with $\hat{s} = 1, \ldots, \bar{s}^M$. Oligopoly states are O_{mn}, with $m = 0, \ldots, \bar{m}$ and $n = 1, \ldots, n_b$, where

$$n_b = \min\{n^*, \bar{n}\}.$$

The numerical bounds are $\bar{s}^M = 30$, $\bar{m} = 50$, and $\bar{n} = 30$. The cutoff n^* is the first follower count at which entrant joining probabilities are numerically zero. At the upper boundaries, monopoly innovation at $\bar{s}^M$, leader innovation at $\bar{m}$, and entrant joining beyond n_b are shut down, while transitions back toward the interior, such as follower catch-up from $\bar{m}$, remain allowed.

For each parameter vector, I first solve the static pricing problem and obtain normalized profits for all states. I then solve for a fixed point in the rate of new-industry creation γ and the neck-and-neck duopoly value v_{01}. Given (γ, v_{01}), the normalized Bellman equations use the effective discount rate $\rho+\gamma$. Holding innovation policies and entrant joining probabilities fixed, I solve the monopoly and oligopoly Bellman equations as linear systems. I then update innovation intensities from the firms' first-order conditions and update entrant joining probabilities from the entrant cutoff rule.

Given the resulting transition rates, I compute the stationary distribution μ over incumbent-industry states. Let $Q(x,\theta)$ denote the transition matrix among incumbent states, and let $b_S(\theta) = z(1-\theta_S)$ denote the rate at which a potential entrant who meets state S creates a new industry rather than joining it. New industries enter state M_1. Thus μ satisfies

$$[Q(x,\theta) - \gamma I + e_{M_1} b(\theta)']\,\mu = 0, \qquad \mathbf{1}'\mu = 1,$$

where e_{M_1} is the unit vector associated with state M_1. The fixed-point condition for γ is

$$\gamma = b(\theta)'\mu = z\left[(1-\theta^M)\sum_{\hat{s}=1}^{\bar{s}^M} \mu^M_{\hat{s}} + \sum_{n=1}^{n_b}\sum_{m=0}^{\bar{m}}(1-\theta_{mn})\mu_{mn}\right].$$

I iterate on (γ, v_{01}) until convergence. After convergence, I verify the solution using Bellman residuals, the normalization and nonnegativity of μ, and the consistency condition $\gamma = b(\theta)'\mu$.

C.2 Internal Calibration and Targeted Moments

The internally calibrated parameters are

$$\Theta = (\sigma, \lambda, \alpha, \psi, \hat{k}, z),$$

where σ is the elasticity of substitution, λ is the innovation step size, α and ψ govern R&D costs, $\hat{k}$ governs the mean market-initiation cost, and z is the arrival rate of potential entrants. For each candidate Θ, I solve the stationary equilibrium and compute model moments $m(\Theta)$. The calibrated parameters solve

$$\hat{\Theta} = \arg\min_{\Theta} \left(m(\Theta) - m^{data}\right)' W \left(m(\Theta) - m^{data}\right),$$

where W is diagonal with $W_{jj} = w_j/(m_j^{data})^2$. Thus each moment enters as a weighted squared proportional deviation from its empirical target. I assign somewhat larger weights to the markup-dispersion and innovation-frequency moments.

The model counterpart of aggregate growth is the finite-state version of Proposition 5.1:

$$g(\Theta) = \ln \lambda \left(\gamma + \sum_{\hat{s}=1}^{\bar{s}^M} \mu_{\hat{s}}^M x_{\hat{s}}^M + \sum_{n=1}^{n_b} \sum_{m=1}^{\bar{m}} n x_{-mn} \mu_{mn} \right).$$

The three terms are variety creation, monopolist innovation, and follower catch-up innovation.

The model R&D-to-GDP ratio includes incumbent R&D and market-initiation costs:

$$\frac{R\&D}{Y} = \sum_{\hat{s}=1}^{\bar{s}^M} \mu_{\hat{s}}^M R(x_{\hat{s}}^M) + \sum_{n=1}^{n_b} \sum_{m=0}^{\bar{m}} \mu_{mn} \left[R(x_{mn}) + nR(x_{-mn}) \right] + z \sum_{S} \mu_S \int_{\underline{\kappa}}^{\hat{\kappa}_S} \kappa \, dG(\kappa),$$

where S indexes industry states and $\hat{\kappa}_S$ is the state-specific cutoff below which a potential entrant creates a new industry.

The model innovation-frequency moment is computed from the aggregate innovation hazard per active firm. Let

$$\bar{F} = \mu^M + \sum_{n=1}^{n_b} \sum_{m=0}^{\bar{m}} (n+1) \mu_{mn}$$

be the average number of active firms per industry. The aggregate innovation hazard per industry is

$$H = \sum_{\hat{s}=1}^{\bar{s}^M} \mu_{\hat{s}}^M x_{\hat{s}}^M + \sum_{n=1}^{n_b} \sum_{m=0}^{\bar{m}} \mu_{mn} \left(x_{mn} + n x_{-mn} \right) + \gamma.$$

The model counterpart of the annual innovation frequency is

$$I^{model} = 1 - \exp\left(-\frac{H}{\bar{F}} \right).$$

The term γ is included because new-industry creation is interpreted as product-market innovation. Leader innovations enter H even though they do not enter the public-knowledge growth decomposition. Dividing by $\bar{F}$ converts the per-industry hazard into a per-firm hazard.

The annual brand-entry moment is

$$E^{model} = \frac{z}{\bar{F}}.$$

This uses the potential-entrant arrival rate z, not the new-industry creation rate γ, because each potential entrant creates a new firm or brand either by joining an existing industry or by creating a new industry. I match this count rate directly to the sales-weighted annual brand-entry rate of 0.085 from Argente, Lee and Moreira (2024), Online Appendix Table B.II.

The markup moments are computed from the model-implied static equilibrium in each industry state and aggregated using the stationary distribution. The model average markup is cost-weighted across active firms and enters the objective net of one. The empirical counterparts are constructed from Compustat over 2000–2016 using the output elasticities of De Loecker, Eeckhout and Unger (2020). For the dispersion moment, I residualize log markups on four-digit-industry and year fixed effects, compute the cost-weighted standard deviation of the residuals within each year, and average across years. This removes permanent cross-industry differences and aggregate year effects, making the target the empirical counterpart of the model's cross-sectional markup dispersion.

C.3 Life-Cycle Profiles

To construct life-cycle profiles, I follow a cohort of industries using the transition process implied by the stationary equilibrium. The active state space is

$$\mathcal{X} = \{M_1, \dots, M_{\bar{s}^M}\} \cup \{O_{mn} : m = 0, \dots, \bar{m},\ n = 1, \dots, n_b\}, \qquad n_b = \min\{n^*, \bar{n}\}.$$

The reported profiles start from the newborn monopoly state M_1. Let $\omega(t)$ denote the distribution of this cohort over active states at age t, conditional on survival. The numerical implementation uses the column-vector convention. The code stores the transpose of the row-vector generator used in the paper, so the cohort law of motion is

$$\dot{\omega}(t) = R\omega(t), \qquad \omega(0) = e_{M_1},$$

where e_{M_1} puts unit mass on M_1. Equivalently, under the row-vector convention, $P(t) = \exp(Qt)$, where $Q = R'$. The matrix R describes transitions among active states for a given industry and is distinct from the stationary Kolmogorov equations for industry shares, which also include birth, exit, and dilution terms. In the computation, I integrate the forward equations using a time step $dt = T/n_t$ and renormalize the cohort distribution at each step.

For state variables that depend only on the current state, the expected age profile is

$$\mathbb{E}[A_t \mid M_1,\ \text{survival to } t] = \sum_{x \in \mathcal{X}} \omega_x(t) A(x).$$

This formula applies directly to the number of firms and innovation rates. For output and prices, the implementation also tracks cumulative productivity growth along the cohort path and applies the corresponding technology adjustment to the static state-level objects.

C.4 Discontinuities in Comparative Statics

The comparative statics in Figures 3 and 4 exhibit several discrete jumps in growth and innovation outcomes as parameters vary. These jumps are a feature of the finite-state implementation and the threshold structure of entry decisions, not evidence of numerical instability.

Entrant decisions are governed by cutoffs. In state S, a potential entrant compares the value of joining the incumbent industry, v^f_{S+}, with the value of creating a new industry, $v^M_1 - \kappa$. The entrant creates a new industry if

$$\kappa \leq \hat{\kappa}_S \equiv v^M_1 - v^f_{S+}.$$

Thus the creation probability is $G(\hat{\kappa}_S)$, after truncating the cutoff at the endpoints of the support of G, and the joining probability is

$$\theta_S = 1 - G(\hat{\kappa}_S).$$

As parameters vary, $\hat{\kappa}_S$ can cross an endpoint of the support of G, moving the entry decision in that state to a corner. In addition, the retained follower count

$$n_b = \min\{n^*, \bar{n}\}$$

can change discretely because n^* is the first integer follower count at which entrant joining probabilities are numerically zero.

These switches affect the rate of new-industry creation,

$$\gamma = z\left[(1-\theta^M)\sum_{\hat{s}}\mu^M_{\hat{s}} + \sum_{n=1}^{n_b}\sum_{m=0}^{\bar{m}}(1-\theta_{mn})\mu_{mn}\right].$$

Because γ determines $g_N = \gamma - \delta$ and the effective discounting term $\rho + \gamma$, changes in γ feed back into firm values, innovation rates, and the stationary distribution.

The comparative static with respect to $\hat{k}$ contains an additional low-$\hat{k}$ nonlinearity. Market-initiation costs are drawn from a transformed beta distribution with fixed support and fixed dispersion, so varying $\hat{k}$ changes both the mean and the shape of the distribution. At low values of $\hat{k}$, the distribution places substantial mass near the lower support. Small changes in $\hat{k}$ can therefore sharply change the mass of low-cost ideas relevant for $G(\hat{\kappa}_S)$, and these distributional changes can be amplified by changes in n_b and by the equilibrium response of the stationary distribution. Tightening convergence tolerances and increasing the follower grid produce similar patterns, so I interpret the low-$\hat{k}$ dip as a shape-induced and threshold-amplified equilibrium nonlinearity.